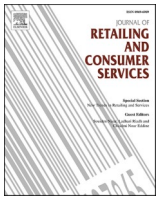
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Review lifecycle analytics and importance–obsolescence analysis: A data-driven design approach to product circularity

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ABSTRACT

Designing products and components with long-term appeal that cater to both early adopters in the new product market and late adopters in the second-hand or remanufacturing markets is essential for extending product longevity and advancing a circular economy through reuse and remanufacturing. Achieving such a circularity-focused design necessitates an understanding of the shifts in customer preferences across different adopter groups, a gap insufficiently addressed in the existing literature. This study seeks to bridge this gap by introducing novel design frameworks called review lifecycle analytics (RLA) and importance-obsolescence analysis (IOA). This systematic review analytics framework reveals dynamic changes in customer preferences over the product adoption lifecycle, enabling design strategies to support product circularity. The RLA-IOA framework analyzes temporal changes in review content through attribute-level sentiment analysis, identifying shifts in the importance and performance of individual attributes between early and late adopters. By generalizing the findings across multiple products, the framework derives the patterns of importance change and obsolescence speed for each attribute, providing valuable design insights for enhancing product longevity through reuse and remanufacturing. As a novel data-driven approach, the RLA-IOA establishes a link between review analytics and sustainable design for circularity. To demonstrate its application and value, this study conducted a case analysis using 63,391 online customer reviews of 12 smartphone models from two brands.

1. Introduction

Product sustainability has emerged as a critical global concern and is a top priority for various industries. Manufacturers, consumers, and governments are becoming increasingly attentive and proactive in addressing these challenges. In this context, a shift toward a circular economy has become imperative, aiming to decouple economic growth from resource consumption and waste generation (Kumar et al., 2021). Unlike the "make-use-dispose" model of the traditional linear economy, the circular economy adopts a "make-use-reuse" approach, extending product longevity through circular business models, such as reuse and remanufacturing (Morseletto, 2020). Reuse and remanufacturing enable products to enter their second life at a fraction of the original production cost and material resource consumption, contributing significantly to environmental sustainability by reducing natural resource depletion, greenhouse gas emissions, and waste generation (Kwak and Kim, 2017; Kurilova-Palisaitiene et al., 2024).

Reflecting this trend, the global circular economy is expanding rapidly. The total number of circular economy transactions in consumer goods, including reuse, remanufacturing, and rental, was estimated to reach approximately 20 billion by 2022 (Statista, 2023a). Global revenue from circular economy transactions was valued at $339 billion in 2022 and is expected to more than double by 2026 (Statista, 2023b). This growth spans multiple industries, including aerospace, automotive, electronics, heavy-duty and off-road equipment, and machinery, where leading companies such as Caterpillar, Cummins, Xerox, Bosch, Apple, and Siemens have successfully leveraged circular business models as both profitable and sustainable strategies (Singhal et al., 2020; Arnold et al., 2021).

With the transition to a circular economy gaining momentum, the demand for design-driven solutions that enhance product longevity and circularity has reached an unprecedented level. Design plays a pivotal role in enhancing the feasibility and profitability of circular business models (Ahmad et al., 2018; Singhal et al., 2020). Product designs with long-term market appeal are essential to ensure a sustainable circular economy.

As Fig. 1 illustrates, the primary target customers differ between the new and second-life (second-hand or remanufactured) product markets:

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0969-6989/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

early adopters dominate the new product market, whereas late adopters drive demand in the subsequent second-life market. Satisfying both adopter groups is crucial because the success of reuse and remanufacturing depends on two key factors: a sufficient supply of end-of-life products and sustained demand for second-life products. This necessitates circular designs that focus on long-term appeal, specifically, market-oriented circular designs that cater to both new and second-life markets (Kwak and Kim, 2017; Pruhs et al., 2024; Ahmed et al., 2024). Optimizing the initial design specifications and upgradability during the early design stage is essential for meeting the needs of both early adopters in the new product market and late adopters in the second-life market (Kwak and Kim, 2013; Sierra-Fontalvo et al., 2024a).

A key challenge in a market-oriented circular design is the dynamic nature of customer preferences. Due to differences in purchase timing and technology adoption preferences, late adopters' needs may not always align with those of early adopters, requiring design foresight (Rogers, 1976). Therefore, a fundamental prerequisite for circular design is a deep understanding of the diverse needs of early and late adopters at different points in time, particularly how their evaluations of the same design differ, an aspect largely overlooked in the existing literature.

Although the preferences of early adopters in the new product market have been studied extensively, the needs of late adopters remain relatively underexplored (Govindan et al., 2019). Despite the growing interest in circular design, many existing approaches fail to account for the distinct preferences of late adopters. Moreover, shifts in preferences across adopter groups have not been thoroughly examined, posing a significant challenge for design evaluation and optimization of circularity.

When evaluating product design from a circular economy perspective (e.g., Saidani et al., 2019; Mesa and González-Quiroga, 2023; Shevchenko et al., 2024a) or optimizing design specifications for circularity (e.g., Kwak and Kim, 2013; Aydin and Badurdeen, 2019), many approaches assume late adopters' demand based on theoretical models or simply presume that their preferences are either predefined or identical to those of early adopters, leading to potential mismatches in product appeal and optimal specifications. This research gap underscores the urgent need to investigate how early and late adopters' customer opinions of the same product differ, ultimately enabling more effective market-oriented circular design strategies. By incorporating these insights, businesses can develop products that maintain market appeal throughout their adoption lifecycle, while supporting circular economy principles.

To address this research gap, this study introduces novel review analytics frameworks called review lifecycle analytics (RLA) and importance-obsolescence analysis (IOA). The proposed RLA-IOA framework identifies shifts in customer preferences between early and late adopters, thereby providing a data-driven basis for establishing circular design strategies. Fig. 2 presents an overview of the proposed framework.

The RLA-IOA framework leverages online reviews to identify changes in preference between early and late adopters. Online reviews provide valuable insights into customer opinions of various product attributes (Qi et al., 2016; Zhu et al., 2025). As reviews accumulate over time—for example, smartphone reviews posted several years after launch—they reveal how customer evaluations of the same product evolve, spanning from early to late adopters. (Park and Kwak, 2020).

To reveal patterns of opinion change, the RLA-IOA framework segments online reviews by adopter group and conducts attribute-level sentiment analyses, focusing on two key aspects. First, the attributes valued by the different adopter groups varied. For example, early adopters in a new product market may prioritize a smartphone's camera quality, whereas late adopters may find it less important (Park and Kwak, 2020; Lee, 2020). Second, products may undergo technological or physical obsolescence (Sierra-Fontalvo et al., 2024b) between launch and remanufacturing, with the obsolescence rates and degrees differing across attributes (Kwak and Kim, 2016). By analyzing changes in attribute importance and obsolescence speed across adopter groups, the RLA-IOA framework addresses the following research questions.

- **RQ 1:** How does the relative importance of an attribute differ between early and late adopters? Does this increase or decrease?
- **RQ 2:** How does the perceived performance of an attribute differ between early and late adopters? Does obsolescence speed vary?
- **RQ 3:** Which design recommendations can enhance long-term appeal in both new and second-life product markets while supporting circular business models?

By addressing these research questions, the RLA-IOA framework establishes a novel link between review analytics and circular design, offering a structured approach for analyzing shifts in customer preferences across adopter groups. By generalizing the patterns of attribute importance and performance changes, the attributes are categorized into four groups based on (1) changes in importance over time and (2) obsolescence speed. The resulting IOA matrix provides useful design guidelines for each attribute group to promote product circularity. For example, the RLA-IOA emphasizes attributes that gain importance over time but experience rapid obsolescence as they may hinder the demand for second-life products. To mitigate this issue, the framework suggests design-for-circularity strategies such as design for upgrading (Kwak and

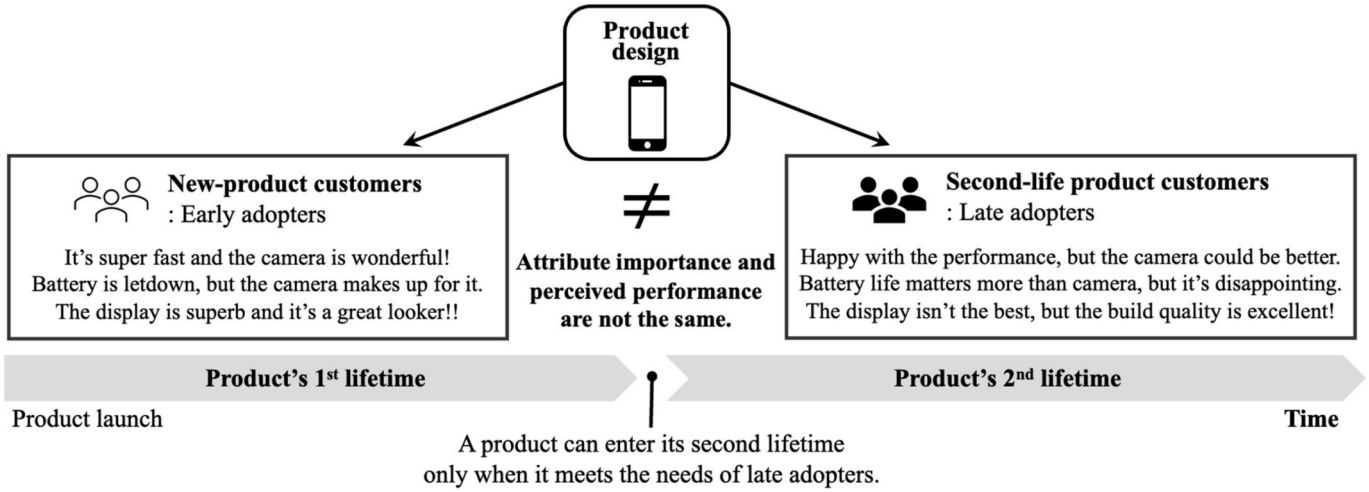


Fig. 1. Motivation of research.

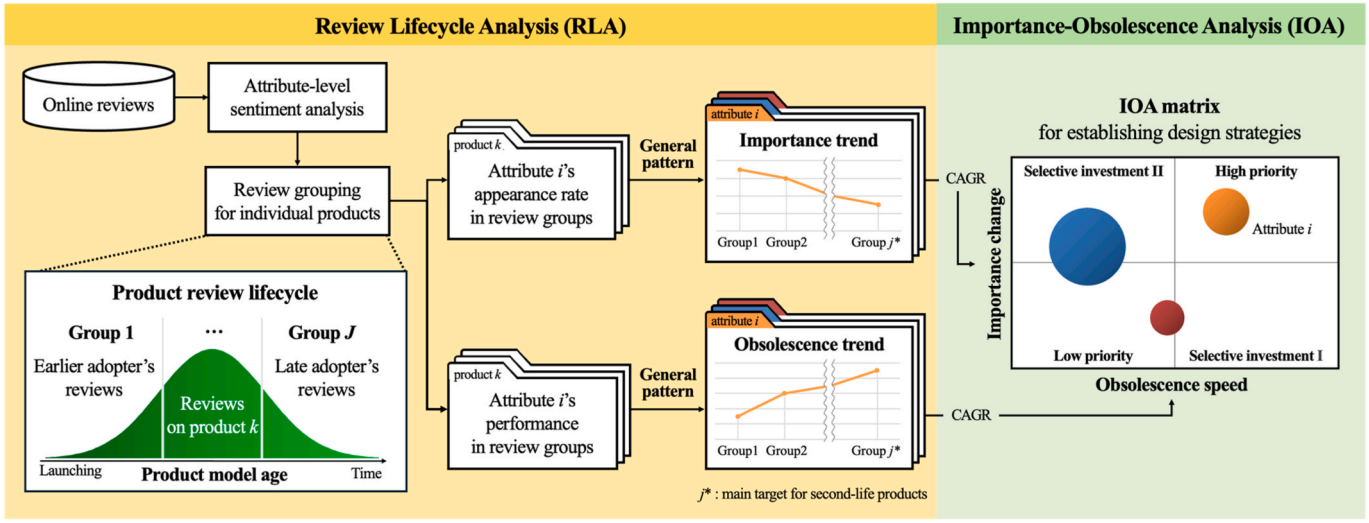


Fig. 2. Overview of the RLA-IOA framework.

Kim, 2013, 2017) or overdesigning (Coman and Ronen, 2010) at the initial design stage.

RLA-IOA is a novel data-driven design tool aimed at promoting product longevity and circularity. This study is one of the first to integrate review analytics with design-for-circularity strategies, potentially inspiring further research and practical applications in fields such as data-driven design, sustainable manufacturing, eco-design, the circular economy, and remanufacturing. To demonstrate the effectiveness of the RLA-IOA framework, this study analyzed 63,391 online customer reviews of 12 smartphone models from two major brands. The remainder of this article is organized as follows: Section 2 reviews related studies; Section 3 describes the RLA-IOA framework; Section 4 presents a case study; Section 5 discusses the implications and contributions of the RLA-IOA; and Section 6 offers conclusions and suggestions for future research.

2. Relevant studies

The RLA-IOA framework is a systematic approach for identifying dynamic customer preferences for circular design purposes and positioning itself at the intersection of the review analytics and design-for-circularity fields. This section reviews two streams of relevant literature: (1) customer preference modeling in the design for circularity, and (2) review analytics for product design. It is important to note that this literature review does not delve into sentiment analysis techniques, such as attribute extraction and sentiment classification, as the primary focus is on integrating review analytics with design-for-circularity strategies rather than advancing sentiment analysis techniques.

2.1. Customer preference modeling in the design for circularity

Understanding customer preferences for second-life products such as remanufactured and refurbished items is essential for making informed design decisions for market-oriented circular designs (Kwak and Kim, 2013; Ahmad et al., 2018; Selvefors et al., 2019; Sharkasi et al., 2025). However, research on how customers value these products, particularly their opinions on the design specifications of remanufactured items, is still in its early stages, which complicates the design-for-circularity decision-making process (Lee and Kwak, 2020).

Most previous studies have relied on customer surveys to explore preferences and purchase intentions in the second-life market and to identify the key factors that influence attitudes toward second-life products. For instance, Wahjudi et al. (2018) and Mugge et al. (2017) analyzed surveys of remanufactured short life-cycle products, such as mobile phones. Milios and Matsumoto (2019) studied the purchase intentions for remanufactured automotive parts using a Swedish consumer survey. Zhang et al. (2019) identified key design factors based on customer requirements through interviews with consumers and stakeholders. Lee and Kwak (2020) examined the factors influencing consumer willingness to pay (WTP) and purchase intention for remanufactured products. Wu et al. (2023) developed a customized remanufacturing method using survey data on individual demands and failure characteristics of second-hand products. Moon (2024) surveyed online flea market users to explore behavioral differences between new and second-hand clothing. Alyahya et al. (2023a, 2023b) developed frameworks to understand the factors influencing consumer decisions to purchase remanufactured and refurbished products.

In addition to surveys, some studies have used empirical research with market data (e.g., sales, price, and end-of-life return data) to identify the factors that influence the value of second-life products. Kwak et al. (2012) analyzed buyback prices for laptops and mobile phones and constructed an assessment model based on product specifications, age, and condition. Zhou and Gupta (2019) examined how innovation levels, product types, and storage conditions affect price differentiation and depreciation rates. In a follow-up study conducted in 2020, Zhou and Gupta explored the impacts of design features and customer attention on these factors (Zhou and Gupta, 2020).

Although these studies provide valuable insights into the factors influencing the market value of second-life products, they fail to address specific challenges in circular design. A successful circular economy requires products that appeal to both early adopters in the new-product market and late adopters in second-life markets. This requires understanding how adopter groups perceive the same product differently. However, the existing approaches do not adequately capture how customer evaluations of a product evolve throughout its adoption lifecycle. Although surveys can provide insights into consumer behavior, they are time-consuming and lack the ability to provide dynamic, real-time information.

2.2. Customer review analytics for product design

Collecting comprehensive market data from early and late adopters is challenging. New solutions are required to address these limitations and bridge this critical gap. One promising approach is to use review analytics, which leverages online reviews to identify preference-changing patterns and develop effective circular design strategies. Online reviews are publicly available, can be easily collected, and offer lower costs and shorter collection times. Consequently, customer review

analytics have emerged as a valuable tool for product design, enabling the examination of customer needs and preferences (Qi et al., 2016; Zhai et al., 2024).

Review analytics involves transforming a corpus of unstructured customer reviews into actionable insights using natural language processing (NLP) techniques. Typically, this includes two steps: text mining and topic modeling to identify keywords related to product attributes, followed by sentiment analysis (or opinion mining) to interpret sentiments toward each attribute (e.g., positive, neutral, or negative) (Lee et al., 2022). Over the past decade, numerous methodological and empirical studies have refined review analytics approaches to derive the design and managerial implications for product improvement. As shown in Table 1, these studies can be categorized based on their research focus (i.e., the variables analyzed) and time considerations.

In terms of research focus, studies fall into three groups: (1) those identifying the relative importance of product attributes; (2) those evaluating product and attribute quality; and (3) those assessing both attribute importance and performance to develop improvement strategies using importance–performance analysis (IPA). Studies that consider time can be grouped into two categories: (1) static models, in which review data are analyzed collectively without considering review timing, and (2) dynamic models, which emphasize time-varying changes in review content to better understand and predict shifts in preferences over time.

Several studies have examined the importance of individual attributes in customer experience and satisfaction. Li et al. (2013) and Ahani et al. (2019) identified the key factors influencing customer satisfaction in hospitality venues through review analytics. Similarly, Kar (2021) investigated the determinants of customer satisfaction with mobile payment, whereas Nasiri and Shokouhyar (2021) explored the critical factors affecting the purchase of refurbished smartphones. Kumar et al. (2023) analyzed the determinants of customer satisfaction with grocery mobile apps using insights derived from online reviews. Güneş (2023) leveraged online reviews to identify new features in coffee machines and immersion blenders, and Miao et al. (2023) applied Latent Dirichlet Allocation (LDA) to extract topics for product design improvement from a customer experience perspective.

Several studies have focused on understanding and predicting changes in attribute importance. By analyzing shifts in attribute importance over time, these studies identified attributes whose relevance was either increasing or decreasing. Timeline analysis of online reviews is commonly used to uncover market-level trends, whereas product-category-level analysis is preferred when reviews of multiple products within the same category are merged and analyzed. For instance, Tucker and Kim (2011) employed statistical trend-detection techniques to forecast future patterns of attribute importance. Jiang et al. (2017) used fuzzy inference and time-series methods to assess and predict the importance of product attributes based on online reviews. Hou et al. (2019) analyzed customer reviews of two successive product generations to track changes in customer expectations and uncover the drivers of these shifts. Martí Bigorra et al. (2019) introduced a Kano categorization method to understand the influence of product attributes on customer satisfaction and the evolution of Kano categories of attributes over time. Similarly, Lee et al. (2022) proposed an importance-Kano analysis that combines review analytics with conjoint analysis to assess the importance of product attributes and performs a dynamic analysis to observe shifts in attribute importance. Zhang et al. (2024) identified important customer requirements at each stage of a

Table 1
Relevant studies on review analytics for product design.

Author	Research focus		Time consideration		Change over adoption lifecycle	Case product
	Importance	Performance	Static	Dynamic		
Li et al. (2013)	●		●			Hotel service
Ahani et al. (2019)	●		●			Hotel service
Kar (2021)	●		●			Mobile payment service
Nasiri and Shokouhyar (2021)	●		●			Refurbished phone
Kumar et al. (2023)	●		●			Grocery mobile app
Güneş (2023)	●		●			Coffee machine, blender
Miao et al. (2023)	●		●			Cordless hairdryer
Tucker and Kim (2011)	●			●		Mobile phone
Jiang et al. (2017)	●			●		Electric iron
Hou et al. (2019)	●			●		E-reader
Martí Bigorra et al. (2019)	●			●		Coffee machine, smartphone
Lee et al. (2022)	●			●		Hotel service
Zhang et al. (2024)	●			●		Mobile phone
Kim et al. (2016)		●	●			Smartphone
Park et al. (2020)		●	●			Hotel service
Lo (2008)		●		●		Community website
Ashton et al. (2015)		●		●		Delivery service
Liang and Wang (2019)		●		●		Water bottle
Joung et al. (2021)		●		●		Uber service
Kim and Lim (2021)		●		●		Mobile game service
Wang et al. (2016)	●	●	●			Hotel reservation app
Nam and Lee (2019)	●	●	●			Airline service
Jeong et al. (2019)	●	●	●			Smartphone
Liu and Zhang (2020)	●	●	●			Hotel service
Joung and Kim (2021)	●	●	●			Smartphone
Chen et al. (2022)	●	●	●			Hotel service
Kim et al. (2025)	●	●	●			Hotel service
Ma et al. (2024)	●	●	●			Food delivery service
Shin et al. (2024)	●	●	●			Hotel service
Wu et al. (2018)	●	●		●		Smartphone
Bi et al. (2019)	●	●		●		Hotel service
Wang et al. (2022)	●	●		●		Urban park
Kim and Kwak (2023)	●	●		●		Hotel service
Jin et al. (2016)	●	●		●	●	Mobile phone
Park and Kwak (2020)	●	●		●	●	Smartphone
Lee (2020)	●	●		●	●	Smartphone
This study	●	●		●	●	Smartphone

product's lifespan: manufacturing, sales and transport, use, and end of life.

Previous studies have focused on assessing the performance of specific products. Product quality and attribute performance are evaluated from the customer's perspective by analyzing sentiment polarity (i.e., whether the sentiment toward an attribute is positive, negative, neutral, or compound) and/or sentiment intensity (i.e., the strength of the sentiment) in each review. Kim et al. (2016) compared the social media opinions of two competing smartphone models and found that social opinions were closely aligned with shipment trends. Park et al. (2020) conducted a comparative study using attribute-level sentiment analysis to identify and compare the competitive advantages of hotel services.

Some studies have emphasized the time-varying changes in attribute performance. Using time-series plots or statistical control charts, these studies have monitored the frequency of negative reviews and average sentiment scores, identified the causes of complaints, and established quality improvement strategies. For instance, Lo (2008) categorized customer complaints and tracked weekly defect rates using control charts. Ashton et al. (2015) enhanced a topic-based quality control scheme to address the limitations of interpreting sudden changes in service quality. Liang and Wang (2019) proposed a joint control chart that detects both topic and sentiment shifts in online customer reviews. Joung et al. (2021) used a review-based approach with time series graphs to identify service failure trends and monitor consecutive failures. Kim and Lim (2021) employed sentiment analysis to identify customer complaints and applied statistical process control charts to monitor complaints over time.

Recent studies have considered both the importance and performance of each product attribute. Importance-performance analysis (IPA), proposed by Martilla and James (1977) remains a widely used tool in this regard. IPA categorizes product attributes based on their relative importance and performance, allowing prioritization for improvement. In general, a four-quadrant graph is used to display importance and performance scores, with each attribute plotted according to its quadrant for strategy suggestions. Although customer surveys have traditionally been used in IPA, recent studies have transitioned to review-based IPAs. Most studies first conducted sentiment analysis to gauge the importance and performance of product attributes, displaying the results in an IPA graph.

Wang et al. (2016) used a review-based IPA to compare two smartphone applications for hotel reservations: online travel agencies and hotel brands. Nam and Lee (2019) employed review-based IPA combined with LDA to define product attributes and related terms. Similarly, Jeong et al. (2019) used LDA and deep-learning-based sentiment analysis for IPA to identify key topics and determine performance scores. Liu and Zhang (2020) conducted a review-based IPA of online hotel reviews, whereas Joung and Kim (2021) employed a deep neural network to obtain more precise importance estimates with a lower variance. Chen et al. (2022) applied a review-based IPA to compare multiple hotels, and Kim et al. (2025) introduced an integrated IPA and Kano model for competitor analysis. Ma et al. (2024) presented a review-based IPA for food delivery services using BERTopic modeling and regression for quality measurement. Shin et al. (2024) proposed a review-based IPA for reliable attribute importance estimation using interpretable machine learning and applied it to a case study of hotel service features in Singapore.

Dynamic IPAs have also been developed to capture the temporal changes in attribute importance and performance. These approaches identify attributes that require urgent improvement based on trends in importance and performance. Wu et al. (2018) applied dynamic IPA to customer review analysis to examine the relative performance of a product's major competitors and attribute significance. Bi et al. (2019) presented various types of review-based IPA with machine-learning techniques to track attribute performance and importance over time. Wang et al. (2022) employed a review-based IPA for urban parks in Beijing, using seasonal changes to identify attributes that require renewal. Kim and Kwak (2023) combined a dynamic IPA with a control chart to monitor complaints and prioritize quality improvement.

In summary, over the past decade, numerous methodological and empirical studies have advanced review analytics techniques to extract design and managerial insights into product improvement. However, a gap remains in the development of systematic frameworks that specifically leverage online reviews for circular design.

Most existing review analytics studies have focused on understanding and predicting broad market trends, namely, which product designs are currently and prospectively preferred by customers. As illustrated in Fig. 3(a), these analyses are typically conducted at the product category level by aggregating the reviews of multiple products within the same category. While useful for identifying general market trends, this approach has two critical limitations: (1) reviewers within the same category group may have different preferences and levels of innovativeness; and (2) product designs within the same category can vary significantly; however, their reviews are analyzed collectively, complicating interpretation and potentially leading to inaccuracies.

2.3. *Review analytics for design-for-circularity strategies: research gap and contributions of this study*

Changes over time in customer preferences for the same product—in other words, differences in design preferences among adopter groups—are critical to circular design but remain underexplored in the existing literature. Notable exceptions include Jin et al. (2016), Park and Kwak (2020), and Lee (2020), who tracked preference changes across the adoption lifecycle of a product. They grouped product reviews by model age (i.e., the time elapsed since their release at the time of the review) and analyzed changes in attribute importance and performance across review groups. Among these three studies, Lee's (2020) approach is the most closely aligned with the RLA-IOA framework. While Jin et al. (2016) and Park and Kwak (2020) focused on attribute trends within a single-product case, Lee (2020) analyzed reviews of multiple products to identify average patterns of change. As illustrated in Fig. 3(b)–Lee (2020) grouped reviews by model age and applied sentiment analysis to explore the evolving trends in attribute importance and performance. By repeating the analysis across multiple products and averaging the results, this study uncovered general patterns of importance and performance changes across review groups.

Jin et al. (2016), Park and Kwak (2020), and Lee (2020) demonstrated that even for the same product, the importance and performance of attributes can shift significantly across the product adoption lifecycle. However, these analyses treated importance and performance as separate dimensions, typically visualizing each through individual line graphs rather than integrating them into a unified framework. Although this approach reveals temporal trends, it lacks a holistic perspective of how these two factors interact dynamically throughout the adoption lifecycle. Furthermore, although these studies confirmed opinion shifts between early and late adopters, they did not provide systematic methods to inform circular design strategies or product design implications.

The RLA-IOA framework introduces a systematic approach for analyzing dynamic changes in customer preferences for circular designs. Specifically, the RLA-IOA clarifies how different adopter groups characterized by varying levels of innovativeness evaluate the same product design differently. The framework applies attribute-level sentiment analysis to track shifts in customer opinions with respect to two key factors: (1) changes in the importance of product attributes and (2) the speed at which these attributes become obsolete. The major product attributes are then categorized into four groups based on changes in importance and obsolescence speed, leading to tailored circular design guidelines.

In summary, the RLA-IOA distinguishes itself by grouping reviews based on the product model age, enabling the discovery of patterns in preference differences between early and late adopters. The framework

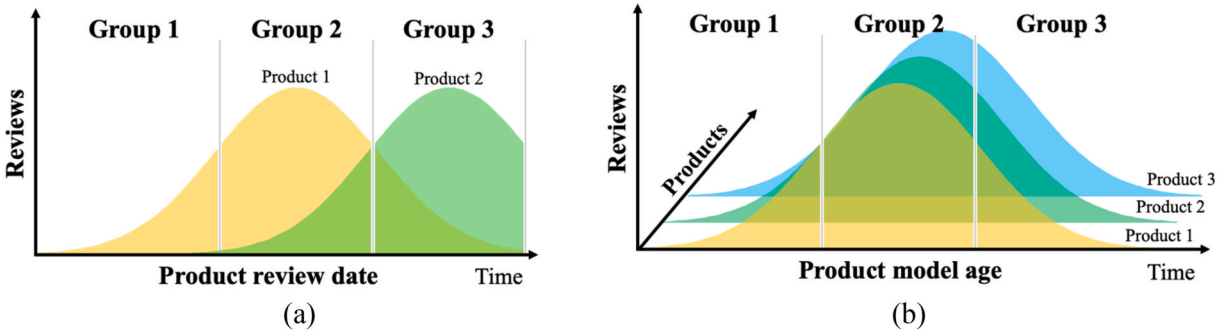


Fig. 3. Review grouping differences: grouping examples in (a) existing approaches; (b) RLA-IOA.

synthesizes data across multiple products while controlling for the effect of product design on review content, and visualizes the results in an IOA matrix for an integrated analysis of importance and performance differences. Although the IOA matrix employs a four-quadrant structure, similar to that of traditional Importance-Performance Analysis (IPA) matrices, it uniquely emphasizes temporal changes between review groups, focusing on the degree and direction of shifts in an attribute's importance and performance from early to late adopters.

The RLA-IOA is one of the first frameworks to facilitate a detailed and holistic analysis of preference differences across the product adoption lifecycle, advancing the development of circularity-oriented design guidelines. By offering a novel data-driven approach, this study bridges the gap between review analytics and sustainable design in circular economies and reports actionable insights to enhance product longevity and support reuse and remanufacturing.

3. Method

This section describes the steps of the RLA-IOA framework. From a circular design perspective, one of the most challenging aspects of achieving long-term market appeal and supporting reuse and remanufacturing lies in addressing attributes that hold relatively greater importance in the second-life market but quickly become obsolete. To address these critical attributes, design remedies, such as design for upgrading or overdesigns, should be implemented at an early stage. The RLA-IOA framework was designed to identify attributes of increasing importance and rapid obsolescence, providing guidance for appropriate circular design strategies that enhance product longevity and circularity.

As illustrated in Fig. 4, the RLA-IOA framework comprises four key steps: (1) attribute-level sentiment analysis, (2) importance analysis, (3) obsolescence analysis, and (4) IOA. Each step is explained in detail in Sections 3.1 through 3.4, respectively, and the nomenclature is presented in Table 2. The framework begins with the application of widely accepted sentiment-analysis techniques to derive attribute importance and performance values from online reviews. The primary contribution of the framework lies in the subsequent steps, in which reviews are grouped based on product age, general patterns of change are extracted across multiple product cases, and a novel matrix is introduced to provide an integrated analysis of attribute importance and performance changes. This enables the development of circular design strategies tailored to each attribute.

3.1. Attribute-level sentiment analysis and review grouping

The first step in the RLA-IOA is attribute-level sentiment analysis, also known as aspect-based sentiment analysis, which extracts opinions on specific attributes from customer reviews and identifies the sentiment polarity (Ray and Singh, 2025). In the RLA-IOA, it is essential to maintain a consistent set of attributes across review groups. Therefore, it is assumed that the key attributes of interest for circular design purposes (i.e., design attributes) are predetermined and provided at the outset.

Although key design attributes are predetermined, they can be referred to in various ways in online reviews. For example, the same attribute might be described as either "display" or "screen" depending on the reviewer. To capture these synonyms or subordinate terms, an attribute lexicon should be created using attribute extraction techniques such as N-grams, LDA, the Apriori algorithm, or Word2Vec. Among these methods, this study employs the deep learning-based Aspect Term Extraction (ATE) model from the PyABSA package (Yang, 2021; Yang et al., 2023), because deep learning algorithms effectively capture the characteristics of word usage by considering both textual context and syntactic information. However, other methods can also be used for the RLA-IOA if they support the required attribute extraction.

A challenge with attribute extraction algorithms is that they extract all words that trigger sentiments in customer review sentences, which may result in the inclusion of irrelevant terms. Therefore, the similarity (relevance) of the extracted terms to the given design attributes must be determined to build an attribute lexicon, comprised only of the relevant terms. Previous research has introduced various methods for constructing attribute lexicons, from manual labeling to unsupervised learning methods such as LDA and K-means clustering. Although unsupervised models allow for automatic grouping, they may not ensure consistency because irrelevant words may be included, or words with different meanings may be grouped. To maintain consistency, this study employed manual labeling, as shown in Fig. 5, to align the extracted words with the predetermined design attributes.

The manual labeling process was as follows: First, words extracted from the entire review dataset were arranged based on their frequency. There is no standardized rule for the extent of labeling. Therefore, researchers can adjust the threshold based on their expertise. Words that appeared more than five times across all review sentences were manually labeled, accounting for approximately 90 % of the total occurrences. The remaining 10 % of occurrences may contain typographical errors, abbreviations, or synonyms. For example, "batteri" is a typo for "battery," and "front cam" is an abbreviation for "front camera." Word embedding, which is an NLP technique that represents words as vectors to capture semantic relationships and similarities, can also be used to label the remaining words. Accordingly, this study used the word embedding model from the FastText package (Joulin et al., 2017) to correct typographical errors and identify synonyms among the unlabeled words.

Word embedding was performed as follows: First, the word-embedding model is trained on online review data to measure the semantic similarity between the extracted words. An unlabeled word is added to the attribute lexicon only if it shows a similarity of 0.8 or higher with a labeled word. The 0.8 threshold, commonly used for correlations, can be adjusted. If the similarity was below 0.8 or no close match was found, the word was deemed irrelevant and excluded. Although manual labeling with word embedding can help lexicons capture diverse linguistic variations, this process is optional. If sufficient manual labeling has already been performed or if the data quality is high, the benefits of word embedding may be marginal, and this step can

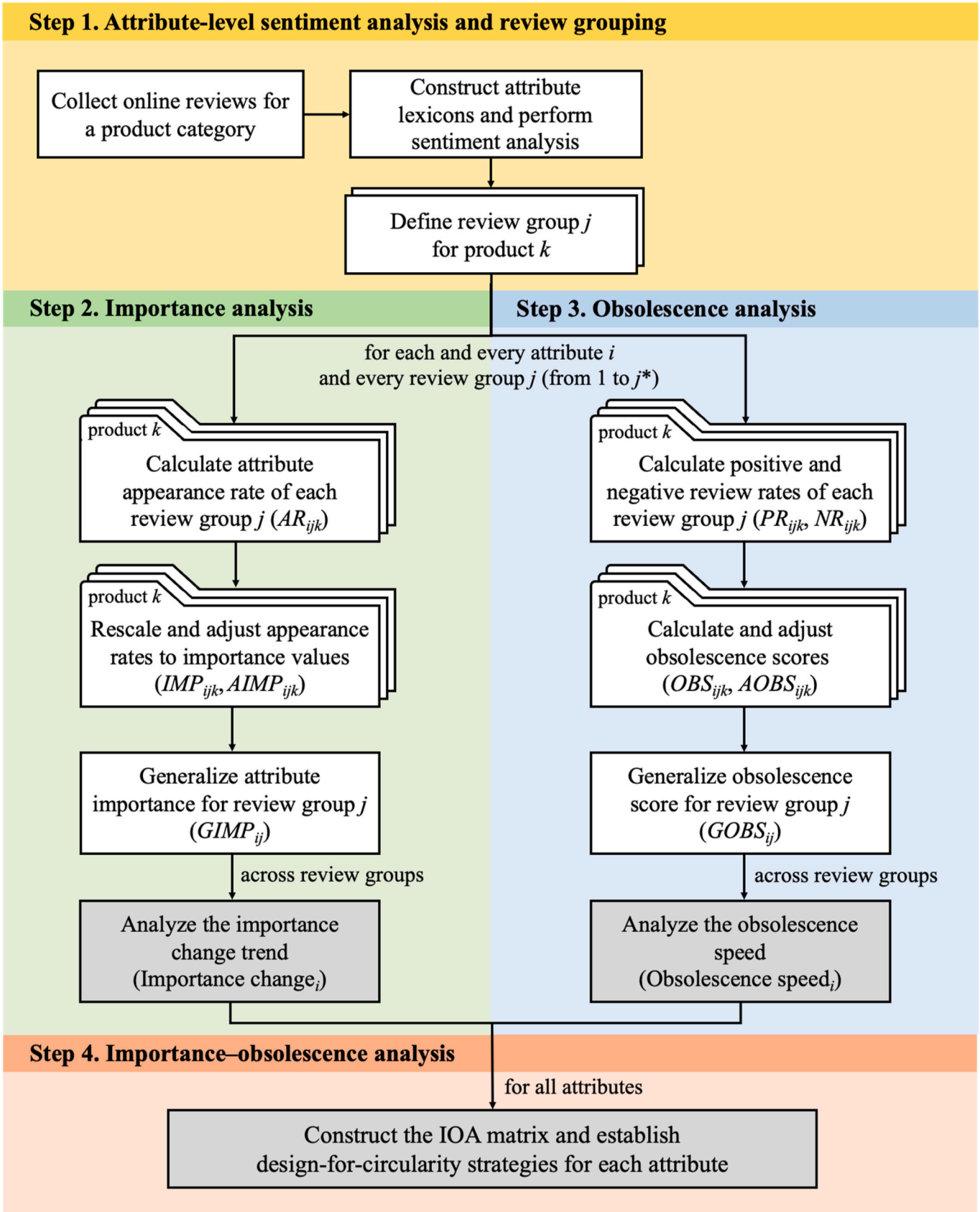


Fig. 4. Procedure of RLA-IOA.

Table 2
Mathematical notations.

Notation	Description
n	Index for reviews of a product, where $n = 1,2,\ldots,N_k$
i	Index for product attributes, where $i = 1,2,\ldots,I$
j	Index for review groups, where $j = 1,2,\ldots,J$
k	Index for products, where $k = 1,2,\ldots,K$
j^*	Target review group representing the main adopter segment for second-life products
X_{in}	Binary indicator of whether attribute i is mentioned in review n
Y_{in}^{pos}	Binary indicator of positive sentiment of attribute i in review n
Y_{in}^{neg}	Binary indicator of negative sentiment of attribute i in review n
AR_{ijk}	Appearance rate of attribute i in review group j for product k
IMP_{ijk}	Rescaled importance of attribute i in review group j for product k
$AIMP_{ijk}$	Adjusted rescaled importance of attribute i in review group j for product k
$GIMP_{ij}$	Generalized importance of attribute i in review group j
Importance change$_i$	Compound annual growth rate of $GIMP_{ij}$ from review group 1 to the target group j^*
PR_{ijk}	Proportion of positive reviews for attribute i in review group j of product k
NR_{ijk}	Proportion of negative reviews for attribute i in review group j of product k
OBS_{ijk}	Obsolescence score for attribute i in review group j for product k
$AOBS_{ijk}$	Adjusted obsolescence score for attribute i in review group j for product k
$GOBS_{ij}$	Generalized obsolescence score for attribute i in review group j
Obsolescence speed$_i$	Compound annual growth rate of $GOBS_{ij}$ from review group 1 to the target group j^*

be skipped.

Once the attribute lexicon is constructed, a sentiment analysis is conducted to predict the sentiment at the attribute level. Previous research primarily employed machine-learning techniques, including support vector machines (SVM) and Logistic Regression, for attribute-based sentiment analysis. However, these approaches have limitations such as not accounting for context and failing to capture phenomena such as sarcasm and neologisms (Lin and He, 2009). In contrast, deep-learning-based sentiment analysis offers a more advanced ability to understand complex contexts by considering word order and relationships within sentences. This allows for a better sentiment analysis of each attribute, even when multiple attributes appear within a single sentence (Sun et al., 2019). Therefore, this study employs a deep-learning-based aspect polarity classification (APC) model from the PyABSA package (Yang, 2021; Yang et al., 2023). Using the previously constructed attribute lexicon, the model identifies the attributes within the review sentences and determines the sentiment polarity (positive, negative, or neutral) of each attribute. Neutral sentiments were excluded, and only the presence of a sentiment for an attribute was considered, without factoring in sentiment intensity or duplication.

The results of the sentiment analysis are presented in Table 3. Review n represents a customer's online review containing multiple sentences. Y_{in} indicates whether the sentiment toward attribute i in review n is positive or negative without considering sentiment intensity or repetition. Neutral sentiments are excluded, and Y_{in}^{pos} and Y_{in}^{neg} denote the sentiment polarity for each attribute i in review n. Even if the sentiment toward an attribute is neutral, the occurrence of the attribute is important for importance analysis. Thus, X_{in} indicates whether attribute i is mentioned in review n regardless of the number of mentions. For instance, if an attribute is mentioned albeit with a neutral sentiment, it is represented as $\{X_{in}, Y_{in}^{pos}, Y_{in}^{neg}\} = \{1,0,0\}$. An attribute can appear with both positive and negative sentiments in a single review and is represented by $\{X_{in}, Y_{in}^{pos}, Y_{in}^{neg}\} = \{1,1,1\}$.

To examine how a product's importance and performance evolve across adopter groups, it is essential to define review groups that represent distinct adopter categories. For each product, review groups are determined based on the product's model age at the time of review, as measured from its release date. The timespan for each group is defined by considering the product release cycle as this cycle significantly influences perceived obsolescence. For instance, in the case of smartphones, which are typically updated and released annually, reviews can be segmented into one-year intervals starting from the release date of each model. Reviews written within the first year of a product's release are categorized into Group 1, representing the opinions of early adopters.

Once the timespan for the review grouping is defined, the total number of review groups, J, is identified for a given corpus of product reviews. For example, J equals four if reviews for a smartphone model have accumulated for four years after its release (Park and Kwak, 2020). Next, among Groups 1 to J, the decision-maker is required to designate the main target group for second-life products, referred to as Group j*. Target group $j \times$ serves as the benchmark model age for remanufactured products when they enter the second-life market; it represents the late adopter group anticipated to purchase used or remanufactured products in the future.

Decision makers can determine $j \times$ by considering multiple factors, such as component reliability, spare part availability, market size and competition, availability of end-of-life products, customer perceptions of second-life products, legal regulations, and other relevant criteria. Notably, a product's average lifespan (replacement cycle length) is a key factor in determining j*, as it typically represents the point at which end-of-life products used as feedstock for circular business models return for a second life. For instance, in the case of smartphones, decision-makers might select $j^* = 3$, considering that the average lifespan of smartphones worldwide from 2013 to 2020 was 28.1 months, according to Statista (2023c). This aligns with the decision maker's plan to release a remanufactured version of the smartphone within two to three years of its initial release (RefurbMe, 2025).

Once the target review group $j \times$ is determined, the RLA-IOA framework analyzes the temporal changes in attribute importance and performance between Groups 1 and j*. By employing review grouping based on the product model age, the RLA provides insights into how evaluations of the same model evolve over time.

3.2. Importance analysis

The next step in the RLA-IOA is the importance analysis, which quantifies the importance level of each attribute in each review group and extracts general trends across the groups. To evaluate an attribute's importance, the RLA-IOA first calculates its appearance rate (AR_{ijk}), as shown in Equation (1). This represents the proportion of reviews mentioning attribute i at least once within review group j for product k.

$$AR_{ijk}(\text{Appearance rate of attribute } i \text{ in review group } j \text{ of product } k) = \frac{\text{Number of reviews referring to attribute } i \text{ in review group } j \text{ of product } k}{\text{Total number of reviews in review group } j \text{ of product } k} \quad (1)$$

Although this appearance rate effectively compares the importance of attributes within a single review group, it may not allow fair comparisons across different review groups because of the inherent differences in review group characteristics. For example, the average lengths

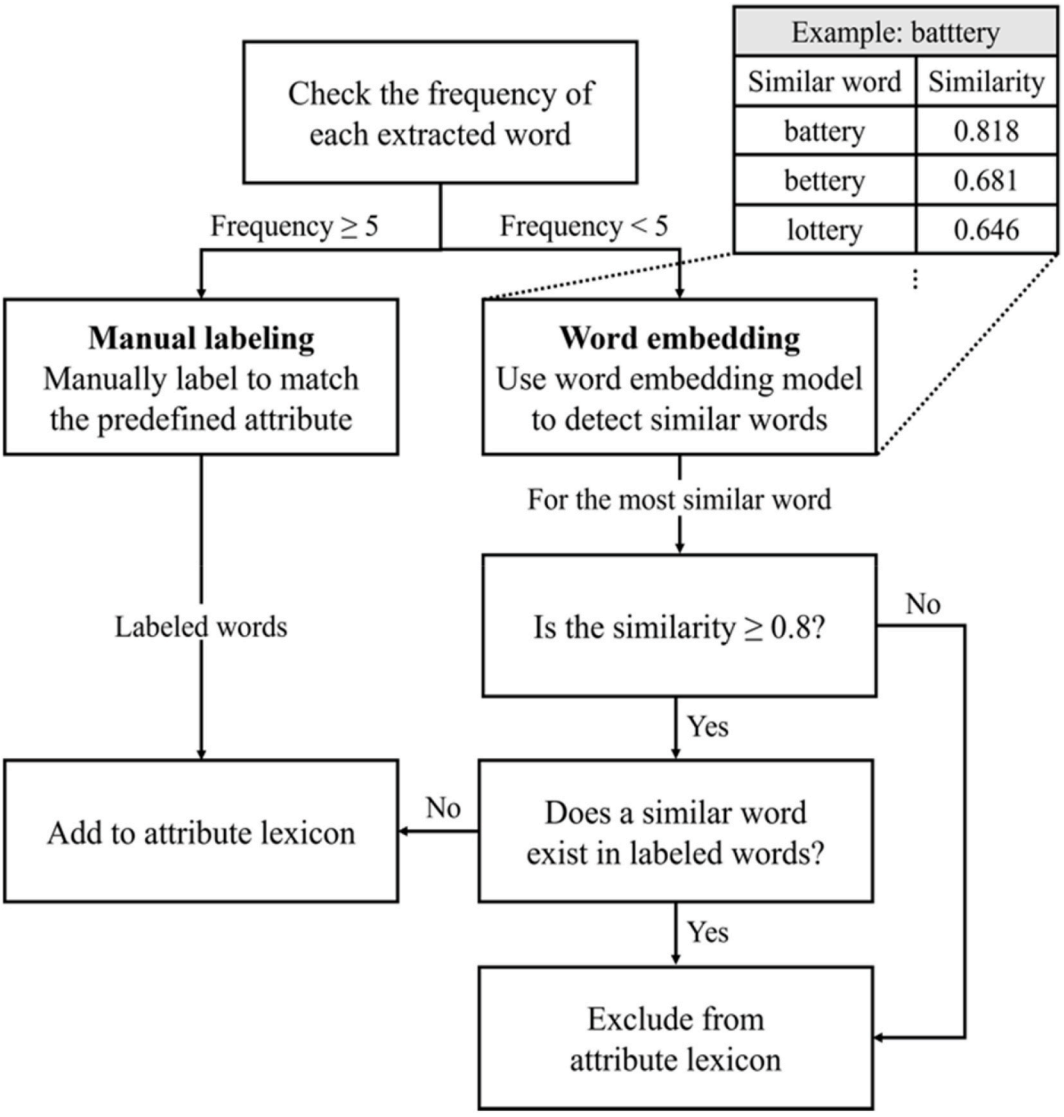


Fig. 5. Manual labeling and word embedding for building an attribute lexicon for RLA-IOA.

Table 3
Data format of sentiment analysis results.

Review n	Attribute 1			…	Attribute i			…	Attribute I		
	X_{1n}	Y_{1n}^{pos}	Y_{1n}^{neg}	…	X_{in}	Y_{in}^{pos}	Y_{in}^{neg}	…	X_{In}	Y_{In}^{pos}	Y_{In}^{neg}
1	1	0	1	…	1	0	0	…	1	1	1
2	1	0	1	…	1	0	1	…	0	0	0
…	…	…		…	…	…		…	…	…	
N_k	0	0	0	…	1	1	1	…	1	1	0

of reviews and customer tendencies to mention attributes can vary across groups (Messner, 2020). Late adopters may mention attributes less frequently because the attribute information is already known, or they may feel that their opinions align with earlier reviews. Therefore, AR_{ijk} cannot be directly compared across review groups because a lower value of AR_{ijk} does not necessarily mean that the attribute is less important. The AR_{ijk} value should be interpreted in relation to other attributes within the same group.

To enable fair comparison across review groups, the RLA-IOA introduces a new metric, IMP_{ijk}, as shown in Equation (2), which rescales each AR_{ijk} by dividing it by the total sum of AR_{ijk} for all attributes within the same group:

$$IMP_{ijk} = \frac{AR_{ijk}}{\sum_i AR_{ijk}} \tag{2}$$

This rescaling ensured that the importance of each attribute was measured relative to the total appearance rate within the same review group, allowing for fair comparisons across groups.

The RLA-IOA also accounts for the fact that IMP_{ijk} can be influenced by the product itself because different products have varying designs and release timings. Because the primary focus of the RLA-IOA is to track how attribute importance changes across review groups for the same product, the product effect must be removed. Therefore, IMP_{ijk} was adjusted to $GIMP_{ij}$, which generalizes the attribute importance by

eliminating the influence of product k. This adjustment occurs in two steps, as shown in Equations (3) and (4):

$$AIMP_{ijk} = \frac{IMP_{ijk}}{IMP_{i1k}} \tag{3}$$

$$GIMP_{ij} = median\left(AIMP_{ij1}, \cdots, AIMP_{ijk}, \cdots, AIMP_{ijK}\right) \tag{4}$$

For each product k, the IMP_{ijk} value for each review group is adjusted to $AIMP_{ijk}$ by dividing by IMP_{i1k}. Naturally, $AIMP_{i1k}$ becomes one for all i and k. The median of all $AIMP_{ijk}$ values across products yields $GIMP_{ij}$, where $GIMP_{i1}$ equals one. Notably, the median was selected as the measure of central tendency owing to the uncertain distributions of $AIMP_{ijk}$, the possibility of a small sample size for k, and the median's robustness against outliers. However, the mean can also be used depending on the decision maker's preference or specific analytical requirements.

The resulting $GIMP_{ij}$ represents the generalized attribute importance for review group j, which can increase or decrease across review groups. To capture the overall importance trend using a single value, RLA-IOA employs the compound annual growth rate (CAGR) of $GIMP_{ij}$, as shown in Equation (5). A positive CAGR indicates that attributes become more important to later customers, whereas a negative CAGR suggests a decline in importance over the product's adoption lifecycle.

$$\text{Importance change}_i = \text{CAGR of } GIMP_{ij} = GIMP_{ij^*}^{\left(\frac{1}{j^*-1}\right)} - 1 \tag{5}$$

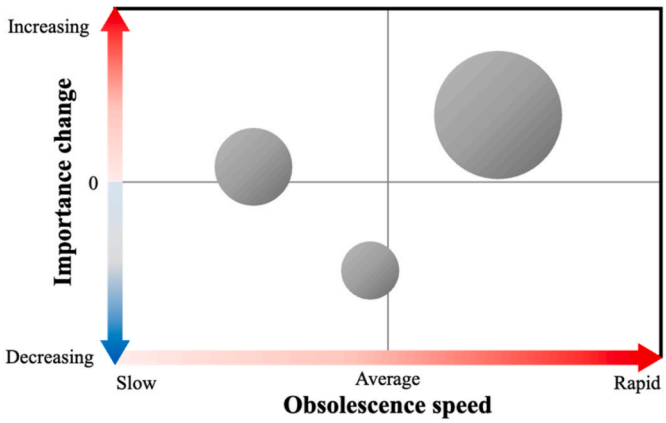


Fig. 6. IOA Matrix: circles represent attributes; size indicates initial importance.

in negative reviews signals a decline in performance, indicating obsolescence. To quantify the degree of obsolescence of each attribute in review group j for product k, the RLA-IOA first evaluates the rates of positive (PR_{ijk}) and negative (NR_{ijk}) reviews, as shown in Equations (6) and (7). The obsolescence score OBS_{ijk} was calculated using Equation (8).

$$PR_{ijk} = \frac{\text{Number of reviews referring to attribute } i \text{ positively in review group } j \text{ of product } k}{\text{Number of reviews referring to attribute } i \text{ in review group } j \text{ of product } k} \tag{6}$$

$$NR_{ijk} = \frac{\text{Number of reviews referring to attribute } i \text{ negatively in review group } j \text{ of product } k}{\text{Number of reviews referring to attribute } i \text{ in review group } j \text{ of product } k} \tag{7}$$

CAGR is a widely used metric that provides a concise summary of overall growth or decline. However, this only accounts for the initial and final values, disregarding intermediate fluctuations. In the RLA-IOA, CAGR is employed as a summary measure under the assumption that the attribute importance changes steadily between the initial and final review groups. Specific patterns, such as U-shaped, inverted U-shaped, or fluctuating trends, revealed by intermediate $GIMP_{ij}$ values should be noted alongside CAGR to capture nuanced variations in attribute importance. More advanced analyses, such as regression or time-series forecasting, can offer deeper insights into the dynamics of change, although they are beyond the scope of this study.

3.3. Obsolescence analysis

Obsolescence refers to the process by which a product or its attributes become outdated and lose utility (Mellal, 2020). This study broadly defines obsolescence as encompassing technological and physical obsolescence. Technological obsolescence occurs when a product becomes outdated due to advances in technology, making older models less desirable or incompatible with new systems. For example, smartphones may become obsolete with the release of newer models with faster processors and improved features.

Physical obsolescence refers to the deterioration of a product's components owing to usage, aging, or environmental factors that reduce its functionality or efficiency. Examples include scratches on screens, rusting of metal parts, and wear and tear of moving components.

The RLA-IOA assesses the degree of obsolescence for each attribute based on sentiment analysis of online reviews, assuming that an increase

$$OBS_{ijk} = \frac{NR_{ijk}}{PR_{ijk} + NR_{ijk}} \tag{8}$$

Similar to the importance analysis, OBS_{ijk} is adjusted and generalized, as shown in Equations (9) and (10). $AOBS_{ijk}$ divides OBS_{ijk} by OSB_{i1k} to account for the differences in product design and inherent performance levels. $GOBS_{ij}$ generalizes the degree of obsolescence across products in review group j using the median $AOBS_{ijk}$ values. Typically, $GOBS_{ij}$ exhibits an increasing trend starting from $GOBS_{i1}$, which is set to one. If unexpected patterns emerge, such as fluctuations and U-shaped or inverted-U-shaped trends, they should be noted, as in the importance analysis.

$$AOBS_{ijk} = \frac{OBS_{ijk}}{OBS_{i1k}} \tag{9}$$

$$GOBS_{ij} = median\left(AOBS_{ij1}, \cdots, AOBS_{ijk}, \cdots, AOBS_{ijK}\right) \tag{10}$$

The obsolescence speed is represented by the CAGR of $GOBS_{ij}$, as shown in Equation (11). A higher CAGR indicates faster obsolescence, whereas a lower value indicates slower obsolescence. By comparing CAGR values, one can identify attributes that are more likely to hinder the reuse or remanufacturing of a product.

$$\text{Obsolescence speed}_i = CAGR\ of\ GOBS_{ij} = GOBS_{ij^*}^{\left(\frac{1}{j^*-1}\right)} - 1 \tag{11}$$

Table 4
Design-for-circularity strategy recommended by RLA-IOA.

Quadrant	Characteristics	Design strategy
I (High Priority)	Greater importance but rapid obsolescence	Depending on the cause of obsolescence, consider: • Technological: Design for upgrades or overdesign • Physical: Durability improvement, design for repair/replacement
II (Selective Investment II)	Increasing importance but slow obsolescence	Selectively invest. Consider increasing durability and reusability to ensure a second life. Exercise caution when investing in technological upgrades.
III (Low Priority)	Decreasing importance but slow obsolescence	If the initial importance is low, avoid overinvestment.
IV (Selective Investment I)	Decreasing importance and rapid obsolescence	For attributes with high initial importance, consider: • Technological: Design for upgrades or overdesign • Physical: Durability improvement, design for repair/replacement

3.4. Importance-obsolescence analysis

Once the importance changes and the obsolescence speed values are obtained, the RLA-IOA integrates the results into an IOA matrix. The IOA matrix is a bubble plot, where the y-axis represents the change in importance and the x-axis represents the obsolescence speed. As shown in Fig. 6, the IOA matrix classifies the product attributes into four groups based on these two factors. The y-axis centerline is 0, which categorizes attributes according to whether their importance is expected to increase or decrease over the product adoption lifecycle. The x-axis centerline was set at the average obsolescence speed of all attributes, distinguishing those experiencing relatively rapid obsolescence from those experiencing relatively slow obsolescence.

Each product attribute is plotted as a bubble in the IOA matrix, where the bubble size represents the historical initial importance of the attribute (i.e., the median of IMP_{i1k} of all k.). In combination with the y-axis value, the bubble size helps estimate how an attribute's importance will change over the adoption lifecycle and highlights which attributes are more likely to become important for second-life products compared to new products.

The IOA matrix recommends design-for-circularity strategies based on the quadrant in which each attribute is located, thereby enabling the integration of customer requirements into the formulation of these strategies. For example, attributes in Quadrant 1 are given the highest priority as they have increasing importance but experience rapid obsolescence. If obsolescence is caused by technological issues, strategies such as design for upgrades or overdesign during the initial design stage are recommended. If obsolescence is caused by physical deterioration, durability improvement or design for repair and replacement is recommended. Similarly, the IOA matrix offers design strategies for each quadrant, as summarized in Table 4.

4. Case illustration

This section presents a case study using smartphones to demonstrate the application and effectiveness of the RLA-IOA framework. Smartphones are among the most actively traded product categories in both the new and second-life markets. According to the International Data Corporation (IDC), the global market for new smartphones continues to grow, reaching 1.24 billion units in 2024, an increase of 6.4 % from 2023. The revenue is projected to reach $485.3 billion in 2025 (IDC, 2025; Statista, 2025a). Moreover, global shipments of remanufactured and used smartphones reached 309.4 million units in 2023, reflecting a 9.5 % increase from 2022. By 2027, shipments are projected to reach 431.1 million units, with a CAGR of 8.8 % from 2022 to 2027 and an estimated market value of nearly $110 billion (IDC, 2024).

Smartphone reviews are often posted several years after the product's release (Park and Kwak, 2020). This study analyzed online customer reviews of 12 smartphone models from two brands, 'A' and 'B,' each represented by six models (A1–A6 and B1–B6). Since the early 2010s, these two brands have consistently ranked first and second in the global smartphone market, respectively. Between 2012 and 2019, they collectively held approximately 35 % of the global smartphone market and 70 % of the U.S. smartphone market (Statista, 2025b). They are also leading original equipment manufacturers (OEMs) in the second-life market, collectively accounting for more than 70 % of global remanufactured smartphone sales (Statista, 2024). The 12 smartphone models selected for this study represented the mainstream series for each brand. By applying the RLA-IOA framework separately to A1–A6 and B1–B6, this study aims to validate the applicability of the framework and its ability to derive consistent results within a product category as well as brand-specific results that reflect their distinct design characteristics.

To ensure a representative dataset, reviews were collected from one of the most influential e-commerce platforms for smartphones and refurbished devices, which serves as a major marketplace for both new and remanufactured smartphones. Each review includes details such as posting date, smartphone model purchased, review content, and star ratings (1–5). While demographic information such as age, gender, and geographic location of reviewers was not available, the platform from which the reviews were collected was predominantly used by U.S. customers, indicating that the data largely reflect customer behavior within these regions. After filtering out non-English reviews and removing duplicates with identical review content, 63,391 valid reviews remained for analysis.

4.1. Sentiment analysis and review grouping

The first step in the RLA-IOA involves attribute-level sentiment analysis and review grouping. A crucial part of this process is defining the key design attributes of interest. Six key attributes were identified for the smartphone case study, based on the approach suggested by Lee (2020): battery, body, camera, display, function, and mainboard. Lexicons were constructed for these six attributes by extracting relevant terms from all reviews. Words that appeared five or more times were manually labeled, whereas the remaining words were automatically

Table 5
Partial attribute lexicons used in the case study.

Key attribute	Description	Relevant terms (examples)	Count
Battery	Terms related to battery capacity, life, consumption, and charge rate	battery_life, battery_health, battery_capacity, battery_performance, battery_power	136
Body	Terms related to overall design and durability	cosmetic_condition, phone_appearance, water_resistance, build_quality, home_button	344
Camera	Terms related to camera pixels, shutter, lenses, and photo quality	camera_lens, photo_quality, front_camera, rear_camera	132
Display	Terms related to display resolution, size, brightness, and touch sensitivity	screen, AMOLED_display, screen_size, LED_screen, display	166
Function	Terms related to sound, microphone, bluetooth, and facial recognition	speaker, microphone, touch_functionality, Bluetooth, audio	273
Mainboard	Terms related to storage, memory, cores, processor, and performance	motherboard, processor, storage, internal_memory, quad_core	126

Table 6
Accuracy of the sentiment analysis.

Attribute	Aspect extraction				Positive sentiment				Negative sentiment			
	Pre.	Recall	F1	Acc.	Pre.	Recall	F1	Acc.	Pre.	Recall	F1	Acc.
Battery	1.00	0.98	0.99	0.99	0.93	0.88	0.90	0.89	0.84	0.92	0.88	0.87
Body	0.90	0.84	0.87	0.89	0.89	0.84	0.87	0.85	0.75	0.87	0.80	0.79
Camera	0.96	0.97	0.97	0.97	0.95	0.86	0.90	0.90	0.78	0.96	0.86	0.86
Display	0.94	0.97	0.96	0.96	0.89	0.91	0.90	0.89	0.83	0.98	0.90	0.89
Function	0.96	0.82	0.88	0.89	0.91	0.80	0.85	0.84	0.90	0.97	0.93	0.92
Mainboard	0.99	0.88	0.93	0.94	0.80	0.87	0.83	0.82	0.65	0.87	0.75	0.80

Note: precision (Pre.), F1-score (F1), and accuracy (Acc.).

classified using a word-embedding process. Table 5 presents some of the attribute lexicons used in the case study, including the definitions of the six attributes and examples of the extracted terms.

Using these attribute lexicons, sentiment analysis was conducted to determine both the presence and frequency of these attributes in review sentences and to assess the sentiment (positive or negative) associated with each attribute. The accuracy of the sentiment analysis applied in this study is presented in Table 6. Accuracy tests were conducted separately for attribute extraction and sentiment classification. A total of 250 reviews were randomly sampled, ensuring that each attribute appeared at least 100 times. The sentiment analysis results were compared with the authors' manual assessments to calculate precision, recall, F1-score, and accuracy for each attribute and sentiment category. The accuracy of attribute extraction ranged from 89 % to 99 %, with an average of 94 %. For sentiment classification, accuracy ranged from 82 % to 90 % for positive sentiment (averaging 87 %) and from 79 % to 92 % for negative sentiment (averaging 86 %).

Following the sentiment analysis, all the reviews were converted into a formatted dataset as shown in Table 3, and review grouping was conducted. For each smartphone model, reviews were divided into annual segments because the average release cycle for smartphone models is typically one year. The target group for remanufactured products was designated as Group 3 ($j^* = 3$), aligning with industry trends: (1) the average global lifespan of smartphones between 2013 and 2020 was approximately two to three years (Statista, 2023c), and (2) the average release date of remanufactured versions fell between 428 and 908 days after the launch of the new product (RefurbMe, 2025). Based on this, three groups (Groups 1–3) were selected for analysis, comprising a total of 47,103 reviews. Here, Group 1 includes reviews posted after the product's release for up to one year (i.e., when the model is one year old), and Group 2 comprises reviews posted between one and two years after the product's release. Table 7 summarizes the review groups for 12 smartphone models. Differences in the average word count and star ratings across these groups suggest potential changes in customer opinions across the product adoption lifecycle. To further investigate these differences at the attribute level, the RLA-IOA conducts an importance and performance analysis based on the sentiment analysis results.

Table 7
Characteristics of review groups.

Smartphone model	Review group	Number of reviews	Percentage of reviews (%)	Posting date	Average word count	Average star rating
Model A1	Group 1	1157	26.59	2012.05–2013.04	138.70	3.91
	Group 2	1461	33.58	2013.05–2014.04	82.07	3.69
	Group 3	1733	39.83	2014.05–2015.04	39.02	3.59
Model A2	Group 1	1191	33.70	2013.04–2014.03	100.96	3.75
	Group 2	1278	36.16	2014.04–2015.03	50.06	3.60
	Group 3	1065	30.14	2015.04–2016.03	47.63	3.32
Model A3	Group 1	2717	42.16	2014.03–2015.02	81.17	4.19
	Group 2	2071	32.13	2015.03–2016.02	57.92	3.87
	Group 3	1657	25.71	2016.03–2017.02	56.73	3.55
Model A4	Group 1	1325	30.47	2015.04–2016.03	97.02	3.80
	Group 2	1194	27.45	2016.04–2017.03	50.61	3.58
	Group 3	1830	42.08	2017.04–2018.03	49.81	3.15
Model A5	Group 1	1122	26.66	2016.03–2017.02	86.62	3.77
	Group 2	1588	37.73	2017.03–2018.02	56.27	3.65
	Group 3	1499	35.61	2018.03–2019.02	56.44	3.31
Model A6	Group 1	264	16.57	2019.03–2020.02	55.02	4.20
	Group 2	544	34.15	2020.03–2021.02	72.31	4.11
	Group 3	785	49.28	2021.03–2022.02	57.13	3.94
Model B1	Group 1	200	8.59	2011.10–2012.09	142.58	3.80
	Group 2	791	33.98	2012.10–2013.09	71.10	3.70
	Group 3	1337	57.43	2013.10–2014.09	57.06	3.58
Model B2	Group 1	542	15.47	2012.09–2013.08	117.77	3.79
	Group 2	771	22.01	2013.09–2014.08	60.32	3.55
	Group 3	2190	62.52	2014.09–2015.08	37.70	3.60
Model B3	Group 1	551	14.13	2013.09–2014.08	79.01	3.85
	Group 2	1791	45.93	2014.09–2015.08	34.54	4.01
	Group 3	1557	39.93	2015.09–2016.08	35.47	3.73
Model B4	Group 1	944	23.52	2014.09–2015.08	57.49	4.26
	Group 2	1124	28.01	2015.09–2016.08	41.58	3.99
	Group 3	1945	48.47	2016.09–2017.08	40.69	3.26
Model B5	Group 1	557	12.30	2015.09–2016.08	65.16	3.90
	Group 2	1509	33.33	2016.09–2017.08	42.93	3.79
	Group 3	2461	54.36	2017.09–2018.08	45.14	3.72
Model B6	Group 1	544	12.50	2016.09–2017.08	62.91	3.63
	Group 2	1107	25.44	2017.09–2018.08	43.17	3.52
	Group 3	2701	62.06	2018.09–2019.08	44.21	3.49

Table 8
Proportion of reviews that referred to each attribute (AR_{ijk}).

Model	Review group	Battery	Body	Camera	Display	Function	Mainboard
Model A1	Group 1	0.21	0.28	0.14	0.33	0.18	0.15
	Group 2	0.17	0.18	0.09	0.16	0.10	0.07
	Group 3	0.11	0.09	0.02	0.06	0.04	0.03
Model A2	Group 1	0.16	0.18	0.16	0.22	0.14	0.11
	Group 2	0.11	0.09	0.06	0.08	0.06	0.05
	Group 3	0.11	0.11	0.04	0.10	0.06	0.04
Model A3	Group 1	0.15	0.17	0.13	0.17	0.13	0.06
	Group 2	0.13	0.17	0.09	0.11	0.09	0.05
	Group 3	0.15	0.17	0.08	0.13	0.11	0.05
Model A4	Group 1	0.22	0.20	0.15	0.15	0.12	0.10
	Group 2	0.16	0.18	0.09	0.10	0.07	0.05
	Group 3	0.21	0.15	0.07	0.10	0.07	0.03
Model A5	Group 1	0.15	0.22	0.17	0.15	0.12	0.08
	Group 2	0.11	0.20	0.07	0.10	0.08	0.04
	Group 3	0.19	0.20	0.05	0.10	0.07	0.04
Model A6	Group 1	0.15	0.31	0.07	0.09	0.13	0.05
	Group 2	0.20	0.32	0.10	0.13	0.14	0.08
	Group 3	0.25	0.28	0.09	0.11	0.16	0.05
Model B1	Group 1	0.14	0.18	0.12	0.13	0.14	0.12
	Group 2	0.07	0.21	0.05	0.07	0.08	0.09
	Group 3	0.09	0.17	0.04	0.06	0.10	0.05
Model B2	Group 1	0.13	0.27	0.11	0.20	0.13	0.11
	Group 2	0.08	0.18	0.05	0.08	0.06	0.05
	Group 3	0.15	0.14	0.02	0.07	0.05	0.02
Model B3	Group 1	0.08	0.21	0.11	0.10	0.15	0.10
	Group 2	0.06	0.12	0.03	0.05	0.05	0.03
	Group 3	0.08	0.14	0.03	0.08	0.06	0.04
Model B4	Group 1	0.06	0.12	0.07	0.09	0.07	0.04
	Group 2	0.04	0.14	0.05	0.08	0.06	0.03
	Group 3	0.12	0.14	0.05	0.12	0.08	0.03
Model B5	Group 1	0.05	0.16	0.09	0.10	0.12	0.09
	Group 2	0.11	0.21	0.04	0.07	0.07	0.03
	Group 3	0.19	0.23	0.05	0.08	0.09	0.03
Model B6	Group 1	0.07	0.17	0.11	0.07	0.11	0.06
	Group 2	0.06	0.19	0.04	0.08	0.08	0.02
	Group 3	0.18	0.20	0.04	0.08	0.11	0.02

4.2. Importance analysis

Importance analysis begins by calculating the appearance rate (AR_{ijk}) for each attribute. The appearance rates were derived for each product model, as listed in Table 8. AR_{ijk} values are useful for comparing attribute importance within a single review group. For example, early adopters (Group 1) of Model A1 prioritized the display and body, while placing less importance on the mainboard and camera. However, directly comparing AR_{ijk} values across review groups is problematic because decreasing word counts across groups (Table 7) imply that a lower appearance rate does not necessarily indicate lower relative importance. To ensure fair comparisons of attribute importance across review groups, AR_{ijk} values must be rescaled to IMP_{ijk}. Table 9 presents the resulting IMP_{ijk}, which reveals the increasing or decreasing patterns across the review groups for the individual models. Fig. 7(a) illustrates the importance of Model A1.

Next, IMP_{ijk} was divided by IMP_{ij1} and adjusted to $AIMP_{ijk}$ to remove the effects of design differences between product models. This adjustment allowed the comparison of $AIMP_{ijk}$ across different models and facilitated the extraction of representative trends across review groups. After calculating $AIMP_{ijk}$ for all models (A1–A6 and B1–B6), general trends were identified for each brand by connecting the median $AIMP_{ijk}$ values (i.e., $GIMP_{ij}$). Fig. 7(b) illustrates the general trend for Brand A based on the median $AIMP_{ijk}$ values of Models A1 to A6.

Table 10 presents the results of the importance analysis. Most attributes followed a monotonically increasing or decreasing trend in $GIMP_{ij}$ values, except for the body and display attributes of Brand B, which exhibited irregular patterns. Noting these irregularities, Importance change$_i$ was computed for each attribute using the CAGR of $GIMP_{ij}$. Both Brands A and B exhibited similar trends, with battery and body attributes gaining importance across review groups, suggesting that late adopters placed greater emphasis on these attributes than early adopters. Consequently, designers should prioritize these attributes when determining product specifications and upgradability to support reuse and remanufacturing. Conversely, other attributes demonstrated a decline in importance across review groups, with the camera and mainboard attributes experiencing the most significant reductions, implying that these features may be less critical for circular design considerations.

4.3. Obsolescence analysis

Obsolescence analysis evaluates the speed at which an attribute becomes obsolete by assessing how quickly negative opinions increase and outweigh the positive opinions. Table 11 presents the rates of positive and negative reviews (PR_{ijk} and NR_{ijk}) for each attribute in individual review groups. Overall, the positive rates decreased across review groups, whereas the negative rates increased for most attributes. This trend suggests a decline in perceived performance over time corresponding to a growing level of obsolescence.

The RLA-IOA measures the degree of obsolescence using obsolescence scores (OBS_{ijk}), which are calculated by dividing NR_{ijk} by the sum of PR_{ijk} and NR_{ijk}. Table 12 presents the OBS_{ijk} values. The obsolescence scores generally exhibited an increasing trend across review groups for most models, indicating that attributes tended to become more obsolete over time. While most cases showed a steady rise in the obsolescence score, there were some exceptions in which the scores stabilized, fluctuated slightly, or rapidly increased in the later groups.

Among these attributes, battery and display consistently showed increasing obsolescence across most models, whereas the other attributes displayed more varied trends. Fig. 8(a) illustrates an example of Model A1, in which the battery, display, function, and mainboard

Table 9
Rescaled attribute importance (IMP_{ijk}).

Model	Review group	Battery	Body	Camera	Display	Function	Mainboard
Model A1	Group 1	0.16	0.22	0.11	0.26	0.14	0.12
	Group 2	0.22	0.23	0.12	0.21	0.13	0.09
	Group 3	0.31	0.26	0.07	0.18	0.11	0.08
Model A2	Group 1	0.17	0.18	0.16	0.23	0.14	0.11
	Group 2	0.25	0.21	0.13	0.18	0.13	0.11
	Group 3	0.24	0.23	0.09	0.21	0.14	0.08
Model A3	Group 1	0.18	0.22	0.16	0.21	0.16	0.07
	Group 2	0.20	0.26	0.14	0.18	0.14	0.08
	Group 3	0.22	0.24	0.12	0.19	0.15	0.08
Model A4	Group 1	0.23	0.22	0.16	0.16	0.13	0.11
	Group 2	0.25	0.28	0.13	0.16	0.10	0.07
	Group 3	0.34	0.23	0.10	0.16	0.11	0.05
Model A5	Group 1	0.17	0.25	0.19	0.17	0.13	0.09
	Group 2	0.19	0.33	0.12	0.17	0.13	0.07
	Group 3	0.29	0.31	0.08	0.15	0.12	0.05
Model A6	Group 1	0.19	0.39	0.09	0.12	0.16	0.07
	Group 2	0.21	0.33	0.11	0.13	0.15	0.08
	Group 3	0.27	0.29	0.10	0.11	0.17	0.06
Model B1	Group 1	0.17	0.22	0.15	0.16	0.16	0.15
	Group 2	0.12	0.37	0.09	0.13	0.14	0.15
	Group 3	0.18	0.34	0.08	0.11	0.20	0.10
Model B2	Group 1	0.13	0.29	0.11	0.21	0.14	0.12
	Group 2	0.15	0.36	0.10	0.16	0.13	0.10
	Group 3	0.33	0.30	0.05	0.14	0.12	0.05
Model B3	Group 1	0.10	0.28	0.14	0.14	0.20	0.13
	Group 2	0.17	0.36	0.08	0.16	0.14	0.09
	Group 3	0.18	0.33	0.06	0.19	0.15	0.09
Model B4	Group 1	0.13	0.27	0.16	0.20	0.15	0.10
	Group 2	0.10	0.35	0.13	0.19	0.15	0.08
	Group 3	0.22	0.26	0.08	0.23	0.15	0.05
Model B5	Group 1	0.08	0.26	0.15	0.17	0.19	0.15
	Group 2	0.21	0.40	0.07	0.14	0.12	0.06
	Group 3	0.29	0.34	0.07	0.12	0.14	0.04
Model B6	Group 1	0.11	0.30	0.18	0.12	0.19	0.09
	Group 2	0.13	0.40	0.09	0.16	0.17	0.05
	Group 3	0.28	0.32	0.06	0.13	0.16	0.04

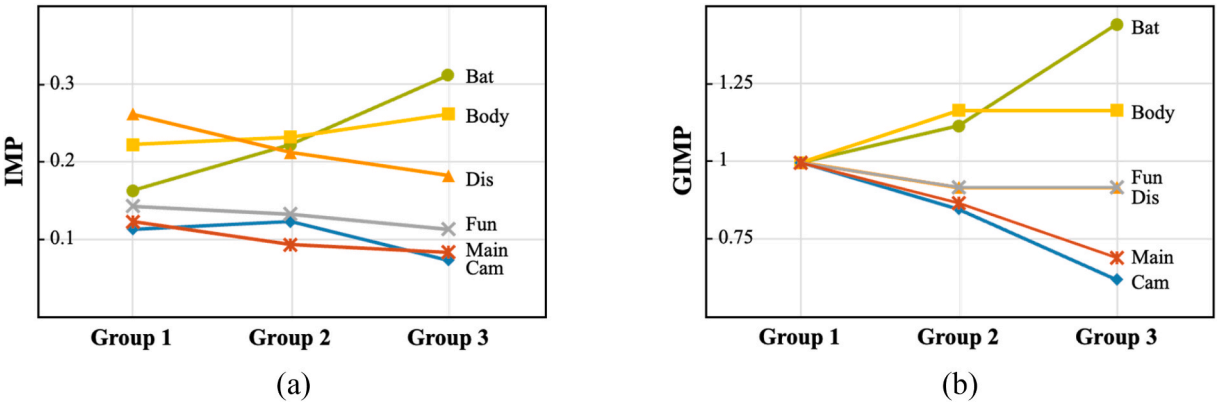


Fig. 7. Importance analysis: (a) rescaled importance of Model A1; (b) generalized importance of Brand A.

Table 10
Importance analysis results: Brands A and B.

Attribute	Brand A				Brand B			
	$GIMP_{ij}$			Importance change$_i$	$GIMP_{ij}$			Importance change$_i$[a]
	Group 1	Group 2	Group 3		Group 1	Group 2	Group 3	
Battery	1.00	1.12	1.45	0.20	1.00	1.11	2.11	0.45
Body	1.00	1.17	1.17	0.08	1.00	1.33	1.14	0.07^{IU}
Camera	1.00	0.85	0.62	−0.21	1.00	0.59	0.46	−0.32
Display	1.00	0.92	0.92	−0.04	1.00	0.88	0.92	-0.04^{U}
Function	1.00	0.92	0.92	−0.04	1.00	0.87	0.85	−0.08
Mainboard	1.00	0.87	0.69	−0.17	1.00	0.75	0.48	−0.31

[a] Note: IU(inverted-U shape); U(U-shape).

Table 11
Proportion of positive and negative reviews that referred to each attribute (PR_{ijk}, NR_{ijk}).

Model	Review group	Battery		Body		Camera		Display		Function		Mainboard	
		PR	NR	PR	NR	PR	NR	PR	NR	PR	NR	PR	NR
Model A1	Group 1	0.48	0.63	0.48	0.65	0.71	0.39	0.70	0.43	0.58	0.54	0.64	0.38
	Group 2	0.28	0.74	0.50	0.55	0.78	0.25	0.67	0.42	0.57	0.48	0.56	0.44
	Group 3	0.12	0.88	0.32	0.71	0.40	0.60	0.28	0.74	0.26	0.73	0.41	0.59
Model A2	Group 1	0.35	0.70	0.49	0.62	0.70	0.37	0.65	0.47	0.62	0.51	0.57	0.53
	Group 2	0.24	0.79	0.31	0.71	0.56	0.51	0.42	0.62	0.30	0.68	0.39	0.64
	Group 3	0.14	0.89	0.38	0.63	0.57	0.46	0.29	0.71	0.17	0.84	0.41	0.49
Model A3	Group 1	0.60	0.45	0.62	0.53	0.76	0.37	0.69	0.38	0.57	0.57	0.61	0.47
	Group 2	0.46	0.57	0.55	0.53	0.63	0.44	0.53	0.50	0.37	0.67	0.46	0.50
	Group 3	0.27	0.69	0.41	0.61	0.48	0.57	0.30	0.73	0.26	0.75	0.42	0.51
Model A4	Group 1	0.37	0.79	0.51	0.60	0.77	0.35	0.66	0.49	0.50	0.63	0.47	0.67
	Group 2	0.21	0.83	0.49	0.56	0.66	0.38	0.40	0.64	0.47	0.56	0.48	0.48
	Group 3	0.11	0.90	0.44	0.60	0.56	0.47	0.19	0.80	0.29	0.74	0.37	0.58
Model A5	Group 1	0.52	0.57	0.49	0.62	0.80	0.25	0.49	0.65	0.54	0.53	0.59	0.48
	Group 2	0.36	0.64	0.51	0.54	0.71	0.35	0.28	0.73	0.38	0.67	0.54	0.38
	Group 3	0.24	0.77	0.43	0.60	0.48	0.48	0.21	0.81	0.34	0.66	0.58	0.45
Model A6	Group 1	0.64	0.33	0.79	0.31	0.67	0.33	0.48	0.52	0.64	0.27	0.57	0.36
	Group 2	0.63	0.36	0.72	0.35	0.82	0.25	0.56	0.48	0.66	0.45	0.81	0.17
	Group 3	0.50	0.52	0.69	0.38	0.68	0.31	0.46	0.54	0.52	0.51	0.76	0.21
Model B1	Group 1	0.43	0.71	0.53	0.58	0.71	0.38	0.65	0.58	0.63	0.52	0.38	0.29
	Group 2	0.24	0.74	0.49	0.50	0.62	0.38	0.39	0.61	0.42	0.60	0.53	0.31
	Group 3	0.16	0.84	0.47	0.57	0.42	0.60	0.26	0.77	0.21	0.79	0.52	0.42
Model B2	Group 1	0.36	0.74	0.54	0.54	0.64	0.47	0.69	0.49	0.48	0.68	0.58	0.37
	Group 2	0.24	0.76	0.48	0.55	0.47	0.53	0.44	0.53	0.58	0.44	0.68	0.25
	Group 3	0.14	0.89	0.40	0.66	0.31	0.69	0.21	0.79	0.19	0.81	0.46	0.42
Model B3	Group 1	0.47	0.65	0.65	0.43	0.64	0.41	0.54	0.63	0.63	0.49	0.45	0.40
	Group 2	0.25	0.74	0.53	0.49	0.57	0.49	0.27	0.75	0.45	0.61	0.41	0.48
	Group 3	0.15	0.84	0.50	0.54	0.50	0.50	0.14	0.87	0.33	0.67	0.46	0.46
Model B4	Group 1	0.48	0.56	0.57	0.58	0.66	0.51	0.53	0.59	0.48	0.63	0.49	0.61
	Group 2	0.50	0.57	0.57	0.51	0.31	0.75	0.42	0.66	0.39	0.64	0.51	0.41
	Group 3	0.15	0.87	0.35	0.68	0.19	0.83	0.10	0.90	0.13	0.88	0.42	0.62
Model B5	Group 1	0.54	0.54	0.66	0.46	0.79	0.37	0.57	0.60	0.69	0.51	0.60	0.36
	Group 2	0.31	0.70	0.60	0.43	0.43	0.62	0.28	0.71	0.38	0.64	0.61	0.39
	Group 3	0.22	0.77	0.61	0.44	0.24	0.79	0.28	0.69	0.28	0.73	0.44	0.49
Model B6	Group 1	0.61	0.50	0.51	0.65	0.71	0.41	0.55	0.55	0.40	0.69	0.77	0.33
	Group 2	0.37	0.68	0.54	0.51	0.49	0.59	0.21	0.76	0.31	0.69	0.56	0.37
	Group 3	0.37	0.62	0.54	0.47	0.32	0.67	0.25	0.72	0.27	0.74	0.44	0.50

indicate increasing trends, whereas the body and camera exhibit fluctuating but generally increasing U-shaped patterns.

To identify a general obsolescence pattern across all the models, OBS_{ijk} values were first adjusted to account for design differences. Rather than focusing on absolute score values, the RLA-IOA emphasizes changes in scores between review groups. To achieve this, each OBS_{ijk} was normalized to $AOBS_{ijk}$ by dividing by OBS_{i1k}. The generalized obsolescence score ($GOBS_{ij}$) was calculated for each review group by taking the median of $AOBS_{ijk}$ values. Fig. 8(b) illustrates an example of $GOBS_{ij}$ for Brand A, where all attributes show increasing trends across the review groups. The detailed results are presented in Table 13. Brands A and B generally exhibited increasing trends, although Brand B showed some U-shaped patterns for the body and mainboard attributes.

Although the general trend shows an increase, the attributes increase at different rates, indicating different rates of obsolescence. In the RLA-IOA, the obsolescence speed of each attribute is summarized as a single number using the CAGR of $GOBS_{ij}$. Table 13 shows the resulting Obsolescence speed$_i$. Higher values indicate a higher rate of obsolescence. For example, the results suggest that displays, cameras, functions, and batteries experience rapid obsolescence, whereas the body and mainboard experience slow obsolescence.

4.4. Importance-obsolescence analysis

Fig. 9 shows the IOA matrices for Brands A and B. In this matrix, each product attribute is represented as a bubble, where the x-axis indicates the speed of obsolescence, the y-axis indicates the change in importance, and the bubble size reflects the initial importance of the attribute. The results revealed both commonalities and distinctions in how product attributes change in importance over time and how quickly they become obsolete.

The importance changes of the attributes exhibit consistent patterns between Brands A and B. The battery and body attributes show increasing importance, whereas the mainboard and camera attributes experience decreasing importance over time. This suggests that late adopters place greater emphasis on the physical durability and battery quality of the device than early adopters, while they are less concerned with processor or camera performance. The display and function attributes are located near the center, meaning that their importance remains relatively stable between early and late adopters, maintaining a moderate level of importance.

Obsolescence speed also revealed overall commonalities between the two brands. For instance, the mainboard and body attributes experience slower obsolescence, implying that their specifications remain sufficient for late adopters. The display and camera attributes exhibited faster obsolescence than other attributes, suggesting that newer models quickly surpass older versions in terms of these attributes, making them more susceptible to obsolescence. Notably, some brand differences were observed. For Brand A, the display attribute showed the highest obsolescence speed (0.34), followed by the camera (0.25). By contrast, for Brand B, the reverse is true: the camera experiences the fastest obsolescence (0.33), followed by the display (0.26). Variations in product innovation strategies, market positioning, and customer preferences and expectations may account for these brand differences.

As outlined in Table 14, the IOA matrix categorizes product attributes into four groups based on the generalized patterns of importance change (increasing or decreasing) and obsolescence speed (relatively fast or slow). Except for battery, all attributes fall into the same quadrant in Brands A and B. For each group, design-for-circularity guidelines are recommended to enhance product longevity and support reuse or

Table 12
Attribute obsolescence (OBS_{ijk}).

Model	Review group	Battery	Body	Camera	Display	Function	Mainboard
Model A1	Group 1	0.57	0.57	0.35	0.38	0.48	0.38
	Group 2	0.73	0.53	0.24	0.39	0.46	0.44
	Group 3	0.88	0.69	0.60	0.73	0.74	0.59
Model A2	Group 1	0.67	0.56	0.35	0.42	0.45	0.48
	Group 2	0.77	0.69	0.48	0.60	0.69	0.62
	Group 3	0.87	0.63	0.45	0.71	0.83	0.54
Model A3	Group 1	0.43	0.46	0.33	0.35	0.50	0.44
	Group 2	0.55	0.49	0.41	0.49	0.64	0.53
	Group 3	0.72	0.60	0.54	0.71	0.74	0.54
Model A4	Group 1	0.68	0.54	0.31	0.42	0.56	0.59
	Group 2	0.80	0.53	0.36	0.62	0.54	0.50
	Group 3	0.89	0.58	0.46	0.81	0.72	0.61
Model A5	Group 1	0.53	0.56	0.24	0.57	0.49	0.45
	Group 2	0.64	0.51	0.33	0.72	0.64	0.42
	Group 3	0.76	0.58	0.50	0.79	0.66	0.44
Model A6	Group 1	0.34	0.28	0.33	0.52	0.30	0.38
	Group 2	0.36	0.33	0.23	0.46	0.41	0.17
	Group 3	0.51	0.35	0.31	0.54	0.50	0.22
Model B1	Group 1	0.63	0.53	0.35	0.47	0.45	0.44
	Group 2	0.75	0.50	0.38	0.61	0.59	0.37
	Group 3	0.84	0.55	0.59	0.75	0.79	0.44
Model B2	Group 1	0.67	0.50	0.42	0.42	0.59	0.39
	Group 2	0.76	0.53	0.53	0.55	0.43	0.27
	Group 3	0.87	0.62	0.69	0.79	0.81	0.48
Model B3	Group 1	0.58	0.40	0.39	0.54	0.44	0.47
	Group 2	0.75	0.48	0.46	0.74	0.58	0.54
	Group 3	0.85	0.52	0.50	0.86	0.67	0.50
Model B4	Group 1	0.54	0.51	0.44	0.53	0.57	0.56
	Group 2	0.53	0.47	0.71	0.61	0.63	0.44
	Group 3	0.86	0.66	0.81	0.90	0.87	0.60
Model B5	Group 1	0.50	0.41	0.32	0.51	0.42	0.38
	Group 2	0.69	0.42	0.59	0.72	0.63	0.39
	Group 3	0.78	0.42	0.77	0.71	0.72	0.53
Model B6	Group 1	0.45	0.56	0.37	0.50	0.63	0.30
	Group 2	0.65	0.48	0.55	0.79	0.69	0.40
	Group 3	0.63	0.46	0.68	0.74	0.74	0.53

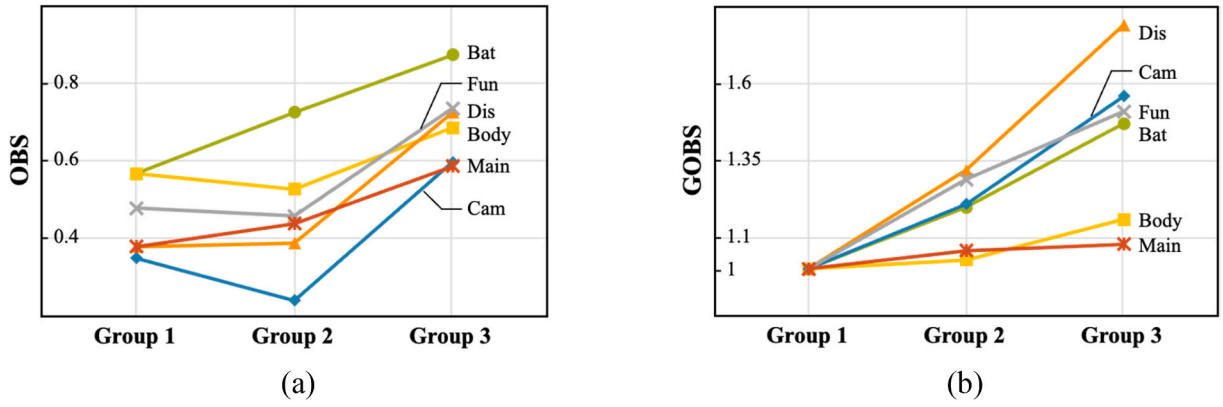


Fig. 8. Obsolescence analysis: (a) obsolescence of Model A1; (b) generalized obsolescence of Brand A.

Table 13
Obsolescence analysis results: Brands A and B.

Attribute	Brand A				Brand B			
	$GOBS_{ij}$			Obsolescence speed$_i$	$GOBS_{ij}$			Obsolescence speed$_i$
	Group 1	Group 2	Group 3		Group 1	Group 2	Group 3	
Battery	1.00	1.20	1.47	0.21	1.00	1.25	1.42	0.19
Body	1.00	1.03	1.16	0.08	1.00	0.99	1.14	0.07 ^{U}
Camera	1.00	1.21	1.56	0.25	1.00	1.36	1.77	0.33
Display	1.00	1.32	1.79	0.34	1.00	1.34	1.60	0.26
Function	1.00	1.29	1.51	0.23	1.00	1.20	1.53	0.24
Mainboard	1.00	1.06	1.08	0.04	1.00	0.94	1.14	0.07 ^{U}

*Note: IU(inverted-U shape); U(U-shape).

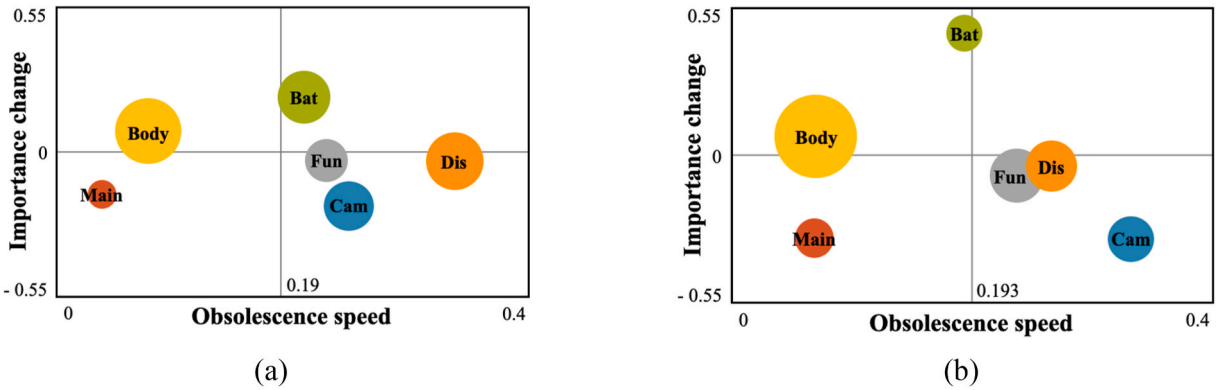


Fig. 9. IOA matrix: (a) Brand A; (b) Brand B.

Table 14
Attributes categorization based on IOA.

Area	Brand A	Brand B
Quadrant I	Battery	–
Quadrant II	Body	Battery, Body
Quadrant III	Mainboard	Mainboard
Quadrant IV	Camera, Function, Display	Camera, Function, Display

remanufacturing. For instance, the battery of Brand A located in Quadrant I, indicating that it is more important for late adopters while experiencing faster obsolescence over the adoption lifecycle. This implies that batteries have a higher priority for improvement before entering their second life. To support product circularity, proactive design improvements should be made depending on the cause of obsolescence. If obsolescence is caused by technological factors, design upgrades or overdesigns during the initial design are recommended. If physical deterioration is the cause, strategies such as improving the durability or designing for easy replacement should be considered.

Selective investment is recommended for attributes in Quadrant II. The body attributes in Quadrant II show increasing importance but slower obsolescence, indicating that technological upgrades are less critical. In contrast, it is advisable to invest in durability and reusability to ensure a second life. The battery attribute of Brand B is also in Quadrant II but exhibits a moderate obsolescence speed close to the center. Thus, investing in technological upgrades or overdesign may be beneficial, as in the case of Brand A.

Attributes in Quadrant IV, such as cameras, functions, and displays, experience faster obsolescence but become less important to late adopters unless they initially hold high importance (represented by a larger bubble size in the plot). For attributes with a high initial importance, selective investment may be warranted to prevent and mitigate obsolescence.

The attributes in Quadrant III, such as the mainboard, were of low priority. These attributes indicate decreasing importance and slow obsolescence. Given the relatively low initial importance of the mainboard, an overdesign or excessive investment beyond the needs of early adopters is unnecessary.

5. Discussion

5.1. Statistical validation of the RLA-IOA results

To validate the RLA-IOA framework and strengthen the interpretation based on its CAGR results, additional statistical analyses can be conducted to test whether attribute importance and performance differ significantly across adopter groups. Non-parametric median tests, such as the Wilcoxon signed-rank test and the sign test, offer a robust approach for assessing whether the median values of Group j^* (i.e., medians of $AIMP_{ij^*k}$ and $AOBS_{ij^*k}$) significantly differ from the median of Group 1 (=1). These statistical tests are particularly useful for handling non-normal distributions, outliers, and small sample sizes (Whitley and Ball, 2022; Nahm, 2016), although each method has limitations and should be applied with caution.

The Wilcoxon signed-rank test assumes a symmetric distribution around the median, making it less reliable for highly skewed data. While the sign test is robust to outliers and does not assume a specific distribution, it may have lower sensitivity in detecting small effects. Bootstrap resampling methods can complement these statistical tests by providing 95 % confidence intervals (CI) for the medians of $AIMP_{ij^*k}$ and $AOBS_{ij^*k}$.

Table 15
Statistical testing and confidence interval estimation for the median of $AIMP_{i3k}$.

Brand	Attribute	$GIMP_{i3}$	Importance change$_i$[b]	Wilcoxon signed-rank test (H_0: $AIMP_{i3k}$ median = 1)		Bootstrap 95 % CI for $AIMP_{i3k}$ median	
				z	p-value	Lower bound	Upper bound
A	Battery	1.45	0.20	2.20	0.0313*	1.31	1.82
	Body	1.17	0.08	0.94	0.4375	0.91	1.24
	Camera	0.62	−0.21	−1.99	0.0625[a]	0.50	0.92
	Display	0.92	−0.04	−1.78	0.0938[a]	0.79	1.00
	Function	0.92	−0.04	−1.57	0.1563	0.84	1.05
	Mainboard	0.69	−0.17	−1.99	0.0625[a]	0.55	0.97
B	Battery	2.11	0.45	2.20	0.0313*	1.39	3.01
	Body	1.14	0.07^{IU}	1.99	0.0625[a]	1.02	1.42
	Camera	0.46	−0.32	−2.20	0.0313*	0.39	0.54
	Display	0.92	-0.04^{U}	−0.31	0.8438	0.70	1.24
	Function	0.85	−0.08	−1.15	0.3125	0.74	1.12
	Mainboard	0.48	−0.31	−2.20	0.0313*	0.34	0.66

[a] p-value≤ 0.1; *p-value ≤0.05.
[b] Note: IU (inverted-U shape), U (U-shape).

Table 16
Statistical testing and confidence interval estimation for the median of $AOBS_{i3k}$.

Brand	Attribute	$GOBS_{i3}$	Obsolescence speed$_i$[b]	Wilcoxon signed-rank test (H_0: $AOBS_{i3k}$ median = 1)		Bootstrap 95 % CI for $AOBS_{i3k}$ median	
				z	p-value	Lower bound	Upper bound
A	Battery	1.47	0.21	2.20	0.0313*	1.30	1.62
	Body	1.16	0.08	2.20	0.0313*	1.06	1.26
	Camera	1.56	0.25	1.99	0.0625[a]	1.11	1.90
	Display	1.79	0.34	2.20	0.0313*	1.21	1.96
	Function	1.51	0.23	2.20	0.0313*	1.32	1.75
	Mainboard	1.08	0.04	0.94	0.4375	0.78	1.40
B	Battery	1.42	0.19	2.20	0.0313*	1.32	1.58
	Body	1.14	0.07 U	1.57	0.1563	0.92	1.29
	Camera	1.77	0.33	2.20	0.0313*	1.46	2.14
	Display	1.60	0.26	2.20	0.0313*	1.43	1.79
	Function	1.53	0.24	2.20	0.0313*	1.27	1.72
	Mainboard	1.14	0.07 U	2.20	0.0313*	1.04	1.58

[a] p-value$\leq$ 0.1; *p-value $\leq$0.05.
[b] Note: IU (inverted-U shape), U (U-shape).

These bootstrap CIs help assess whether the median significantly deviates from 1 by providing a range within which the true median is likely to fall (Mokhtar et al., 2023). The combination of statistical tests and confidence interval estimation supports a more rigorous analysis, ensuring that the observed differences between adopter groups are not merely due to random variations. These additional statistical methods strengthen the reliability of CAGR-based findings by confirming the statistical significance of median differences between adopter groups.

Tables 15 and 16 show the results of additional statistical analyses conducted for the attribute importance and performance, respectively. In this study, Wilcoxon signed-rank tests were conducted on $AIMP_{ijk}$ and $AOBS_{ijk}$ values of Group 3 using R (version 4.3.0) to determine whether the median differed significantly from one (i.e., the medians of $AIMP_{i1k}$ and $AOBS_{i1k}$). Most of the data did not exhibit severe asymmetry, indicating that applying the Wilcoxon signed-rank test was generally appropriate. Given the small sample size (six per attribute), exact tests were performed, with the null hypothesis stating that the median is equal to one.

As shown in Table 15, the importance changes are statistically significant at the 10 % significance level ($\alpha = 0.1$), except for the body and function attributes of Brand A, as well as the display and function attributes of Brand B, which exhibit marginal importance changes (0.08, −0.04, −0.04 and −0.08, respectively). To further complement this test, a bootstrap 95 % CI for the median of $AIMP_{i3k}$ was derived from 1000 resampled datasets, providing additional support for the same conclusion.

Similarly, Wilcoxon signed-rank tests applied to $AOBS_{ijk}$ of Group 3 also support the CAGR-based findings at the 10 % significance level in most cases (Table 16), except for the mainboard attribute of Brand A and the body attribute of Brand B, which exhibit the slowest obsolescence speeds (0.04 and 0.07, respectively). Additionally, the bootstrap 95 % CI for the median of $AOBS_{i3k}$, derived from 1000 resampled datasets, supports the same conclusion.

Although the small sample size limits statistical power and reduces the ability to detect significant differences, the results in Tables 15 and 16 confirm that the interpretation based on the CAGR of $GIMP_{ij}$ and $GOBS_{ij}$ is generally supported by statistical evidence, thereby reinforcing the reliability and robustness of the proposed framework.

5.2. Comparison with conventional approaches

To highlight the distinctiveness and value of the RLA-IOA framework, a comparative study of other methods was conducted, as shown in Fig. 10. Fig. 10(a) presents a dynamic IPA for Brand A created using conventional review grouping based on review dates from 2014 to 2016. For each product model available in a particular year (e.g., Models A1–A3 in 2014, A2–A4 in 2015, and A3–A5 in 2016), the median importance and performance levels of each attribute were calculated and plotted in a dynamic IPA matrix. The two middle lines represent the averages of all the points on the matrix. This dynamic IPA approach reveals general customer opinions toward Brand A products during the given periods, and how these opinions changed over time.

Overall, the attribute importance remained relatively stable during this period, whereas performance showed more significant changes. For example, later models demonstrated improved perceived performance for attributes such as functions, cameras, and mainboards. In contrast, attributes such as display, battery, and body initially improved between 2014 and 2015 but declined between 2015 and 2016, suggesting the

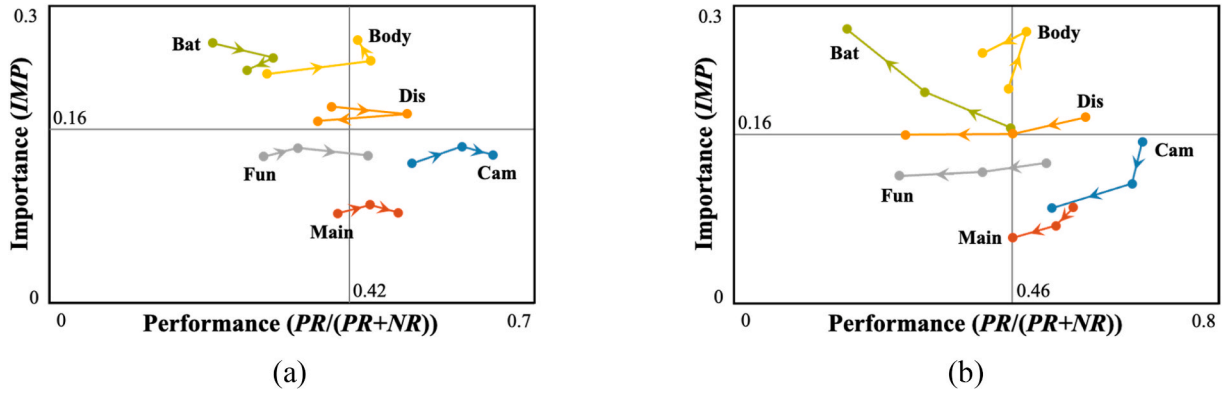


Fig. 10. Comparative analysis for Brand A: (a) dynamic IPA with conventional review grouping (2014–2016); (b) dynamic IPA with review grouping based on product model age.

need for further analysis to understand the causes of these performance drops.

A dynamic IPA with conventional review grouping effectively identifies general market trends and highlights the strengths and weaknesses of a product line coexisting in the market. When determining the initial design of next-generation models, this approach can provide useful insights into the improvement priorities among attributes. Attributes with higher importance but lower performance, such as batteries and displays, emerged as key areas for improvement. However, this approach does not offer insight into the differences between early and late adopters, thereby limiting its utility in circularity-focused design strategies.

Fig. 10(b) demonstrates how review grouping based on the product model age shows different trends. The dynamic IPA matrix plots the median importance and performance levels of each attribute for all product models (A1–A6) across Groups 1–3. Unlike conventional review grouping, this approach revealed decreasing trends in attribute performance, indicating obsolescence. Simultaneously, the attribute importance exhibited more diverse trends, with some attributes increasing in importance and others decreasing over time.

Dynamic IPA with new review grouping is particularly useful for analyzing the strengths and weaknesses of past models from the perspectives of various adopter groups. For instance, battery design has emerged as a critical issue, exhibiting high importance but low performance relative to other attributes. In addition, changes in importance and performance suggest that batteries have become even more critical for late adopters than for early adopters. By contrast, attributes such as cameras and mainboards were less critical for past models, as their high initial specifications mitigated the impact of rapid obsolescence.

The dynamic IPA in Fig. 10(b) is similar to the RLA-IOA framework in that it provides insights into the temporal changes in attribute importance and performance. However, its conclusions relied on absolute performance scores, that were influenced by the designs of past models. This "product effect" complicates efforts to isolate an attribute's inherent characteristics and to understand or compare general patterns of change across adopter groups. For design-for-circularity strategies, the primary focus should be on how the initial performance scores change throughout the product adoption lifecycle rather than on the absolute performance levels of past models. This requirement was addressed using the RLA-IOA framework.

As illustrated in Fig. 9(a), the RLA-IOA emphasizes the inherent characteristics of an attribute by eliminating product effects. For instance, the criticality of the camera was assessed to be relatively higher in Fig. 9(a) than in Fig. 10(b), underscoring the unique insights provided by the RLA-IOA framework. By focusing on changes in importance and performance levels, the RLA-IOA offers clearer and more actionable insights into design-for-circularity strategies, as listed in Table 4.

Table 17
Summary of findings for research questions (RQ1–RQ3).

Research question		Findings
RQ1	How does the relative importance of an attribute differ between early and late adopters? Does it increase or decrease?	The relative importance of attributes shifts between early and late adopters, with some increasing and others decreasing.
RQ2	How does the perceived performance of an attribute differ between early and late adopters? Does obsolescence speed vary?	Perceived performance varies across review groups. The speed of obsolescence varies by attribute.
RQ3	Which design recommendations can enhance long-term appeal in both new and remanufactured product markets while supporting circular business models?	Attributes with rising importance and rapid obsolescence should be the top priority for circular design strategies, such as overdesigning, design for repair, and design for upgrading.

5.3. Implications and contributions of the RLA-IOA framework

The proposed RLA-IOA framework is a data-driven design support tool that identifies dynamic changes in the importance and performance of product attributes throughout the product adoption lifecycle. By combining the sentiment analysis results across distinct review groups, the RLA-IOA uncovers the key dynamic characteristics of product attributes. It captures how customer preferences for a product evolve across adopter groups and how quickly specific attributes become obsolete or unsatisfactory.

Although the case study focused solely on smartphone models, it effectively demonstrated the ability of the RLA-IOA to answer RQ1–RQ3, emphasizing its value and broad applicability. Table 17 summarizes the findings for each research question.

First, regarding RQ1, the case study revealed that (1) the relative importance of an attribute can shift significantly between early and late adopters and (2) these changes vary by attribute, with some increasing in importance and others decreasing. This highlights the critical need to recognize and adapt to these changes when designing products for circularity and marketing second-hand or remanufactured products. By identifying shifts in importance, the RLA-IOA provides businesses with actionable insights to develop design and marketing strategies that reflect evolving customer preferences throughout the product adoption lifecycle.

For RQ2, the case study demonstrated that the perceptions of attribute performance differed between the review groups. Interestingly, performance exhibited significant variation in the speed of obsolescence across attributes. This finding underscores the importance of tailoring design strategies to individual attributes rather than applying a uniform approach. Some attributes may rapidly become obsolete, whereas others may maintain their performance over a longer period. The RLA-IOA enables businesses to create more targeted design solutions that account for varying rates of obsolescence.

RQ3 emphasizes the importance of design strategies that satisfy diverse customer groups across the product adoption lifecycle. In particular, late adopters, who are more likely to purchase second-hand or remanufactured products, play a pivotal role in advancing the circular economy. Understanding their preferences is essential for creating products with long-term appeal. Attributes with high importance and rapid obsolescence are especially relevant in this regard, and the RLA-IOA helps to identify these critical attributes.

For attributes deemed critical to late adopters, the RLA-IOA recommends appropriate strategies such as overdesigning to ensure continued performance despite obsolescence, thereby satisfying both early and late adopters. In addition, designing for repair and upgrading by making parts easier to replace or upgrade during the remanufacturing stage further extends product longevity and enhances circularity. These strategies help products retain their value, continue their second life, reduce waste, and promote sustainability.

The RLA-IOA is one of the first frameworks to bridge the gap between two seemingly distinct fields: review analytics and sustainable design for circularity. By offering a novel and systematic approach to data-driven design for circularity, the RLA-IOA has the potential to inspire new research and practical applications in fields such as data-driven design, sustainable product development, the circular economy, and remanufacturing.

Theoretically, the RLA-IOA makes a novel contribution to the field of data-driven design by advancing the theoretical understanding of how product design influences customer satisfaction in a dynamic context. A common approach in review analytics for product design involves aggregating and analyzing reviews without accounting for the time or context in which they are written. Although some studies have started incorporating time factors, the RLA-IOA distinguishes itself by grouping reviews based on the product's model age, rather than the actual review date. This approach uncovers the preference-shifting patterns between early and late adopters, providing more targeted and actionable insights

into design-for-circularity strategies than merely capturing general market trends.

Moreover, several existing studies have analyzed multiple product reviews but have often overlooked the effect of product design on review content, which can lead to inaccurate interpretations. The RLA-IOA addresses this limitation by emphasizing the need to control for design variations and remove product effects. It recommends rescaling and adjusting product-specific analysis results before integrating data across multiple products to identify general patterns. This nuanced approach reduces the risk of misinterpretation and improves the accuracy of the insights derived from review data.

The RLA-IOA framework also contributes to sustainable design and design-for-circularity research by systematically fusing dynamic customer preferences with circular design. Recent studies on circular design have focused on product-level circularity metric-driven tools (Saidani et al., 2019; Mesa and González-Quiroga, 2023; Shevchenko et al., 2024a) to evaluate product design from a circular economy perspective and implement design-for-circularity strategies (Blomsma et al., 2019; Ruiz-Pastor et al., 2022; Shevchenko et al., 2024b). However, these approaches focus primarily on material-related aspects and often assume that customer needs are static or given. Some studies have aimed to optimize design specifications and/or selling prices to achieve more profitable and environmentally sustainable circular businesses (Kwak and Kim, 2013, 2017; Aydin and Badurdeen, 2019; Hapuwatte et al., 2022). However, these approaches focus mainly on design optimization, in which customer preferences are assumed based on theoretical or predefined demand models. This study distinguishes itself by addressing the dynamic preferences in new and second-hand markets using a data-driven methodology. By complementing other design-for-circularity approaches, this study is expected to help products retain value for longer periods and successfully enter the second-hand market, enhancing not only environmental sustainability but also the economic viability and profitability of circular business models.

Businesses can leverage the RLA-IOA framework to develop data-driven strategies for product improvement, marketing, and sustainable management. By identifying attributes that remain important to late adopters but experience rapid obsolescence, companies can make informed decisions on where to invest in design improvements. The ability of the RLA-IOA to provide actionable insights for enhancing product circularity makes it a valuable tool for industries aiming to promote business sustainability, comply with legislative and regulatory requirements for eco-design and recycling, and explore new opportunities in second-hand markets. As sustainability has become an increasing priority across industries, the proactive approach offered by the RLA-IOA can play a pivotal role in developing more durable, adaptable, and enduringly appealing products, thereby reducing premature obsolescence and minimizing the early disposal of functional products, while facilitating the transition to a more sustainable and circular economy.

From a marketing perspective, the RLA-IOA framework enables businesses to refine messaging for different adopter groups and to develop targeted marketing strategies that emphasize the durability and/or upgraded functionality of specific attributes. Additionally, these insights can be leveraged to identify the primary target segment for second-life products when a product design has already been established. When the initial design specifications are predetermined, the optimal timing for remanufacturing can be determined by simultaneously considering the obsolescence speed, importance, and associated costs of repairs and upgrades.

From a policy standpoint, the insights derived from the RLA-IOA framework can inform regulatory measures that promote sustainable consumption and production. Policymakers can apply this framework to develop data-driven policies that encourage the design of long-lasting, upgradable, and easily repairable products aligned with Extended Producer Responsibility (EPR) regulations and circular economy initiatives. Additionally, the RLA-IOA framework can aid in establishing eco-labeling standards by identifying product attributes that hold the highest value in second-life markets. By incorporating data-driven methodologies such as the RLA-IOA into policy frameworks, governments, and regulatory bodies can foster sustainable production and consumption, accelerating the transition to a resource-efficient net-zero economy.

6. Conclusion

This study proposed the RLA-IOA framework, a novel design-analytics approach aimed at enhancing product circularity. This framework was designed to uncover preference differences between early and late adopters, critical for establishing effective design strategies for circular business models, such as reuse and remanufacturing.

The RLA-IOA framework comprised four key components: (1) attribute-level sentiment analysis, (2) importance analysis, (3) obsolescence analysis, and (4) IOA. Beginning with review grouping based on the product model age at the time of review, the RLA-IOA performed sentiment analysis at the attribute level to assess the importance and obsolescence of individual attributes. Importance was derived from adjusted appearance rates, reflecting general patterns of change over time. Attribute performance was evaluated using positive and negative review rates, resulting in an obsolescence score. This score captured the performance shifts of attributes across different adopter groups, enabling the identification of obsolescence patterns. The IOA matrix, a key interpretive tool, combined the importance and obsolescence analyses to reveal the differences in customer preferences between early and late adopters. This integration facilitated the identification of attributes with high importance but rapid obsolescence, thereby guiding design-for-circularity strategies tailored to each attribute.

To the best of our knowledge, the RLA is the first systematic review analytics framework that incorporates the concept of the product adoption lifecycle and links customer review analytics to circular design strategies. This framework advances our understanding of the dynamic relationship between product design and customer satisfaction in a circular economy, moving beyond static, one-time assessments that typically focus on early adopters in a new product market. The framework plays a pivotal role in supporting circularity and sustainability initiatives by helping companies identify attributes that remain important to late adopters but become quickly obsolete. This approach allows businesses to implement targeted strategies to extend product longevity, such as designing products that can sustain long-term appeal and that can be easily repaired or upgraded. By fostering sustainable consumption practices, the RLA-IOA can contribute to reducing environmental impacts.

Although the case study focused on 12 smartphone models from two major brands, the broader implications of the RLA-IOA extend beyond this single product category. The results revealed common patterns of preference change, underscoring the practical applications of the RLA-IOA and its contributions to sustainable design and the circular economy. To further generalize the findings, future research should apply and validate the RLA-IOA framework across additional smartphone models, brands, and other product categories, such as automobiles, equipment, home appliances, and fashion. This would help identify critical product features and industry-specific circular design strategies, while also assessing the adaptability of RLA-IOA across different market contexts.

Further advancements in each step of the RLA-IOA will be possible through the incorporation of advanced methodologies. For example, although this study constructed an attribute lexicon using researcher-defined keywords, it could be enhanced by employing more sophisticated methods to detect both existing and emerging keywords, resulting in more comprehensive and accurate attribute lexicons. Although the sentiment intensity was not considered in this study, future research could improve the accuracy and reliability of the RLA-IOA by incorporating advanced deep learning models for precise sentiment intensity prediction.

This study employed median- and CAGR-based trend analyses to extract general patterns across multiple product models. Although the median was robust against outliers, it did not fully capture the distribution of all the models. Similarly, CAGR-based analysis focuses solely on the differences between early and late adopters, potentially overlooking intermediate trends. Depending on the characteristics and volume of the data, it may be beneficial to use the mean or adopt alternative statistical methodologies that capture dynamic changes in reviews more effectively. Additionally, a comparative study using surveys on various adopter groups could enhance the reliability of the importance change and obsolescence speed metrics. Furthermore, these metrics may be influenced by broader market trends and may evolve over time. Therefore, they should be continuously updated by incorporating follow-up models.

In particular, this study has inherent limitations regarding the reliability and representativeness of online reviews, which is a common challenge in review-based studies. The RLA-IOA framework requires that a sufficient number of product reviews be accumulated throughout the adoption lifecycle; however, in certain cases, online reviews may be insufficient or unavailable. Additionally, potential biases in the reviews are unavoidable, necessitating careful interpretation. In practice, only a small fraction of customers post reviews relative to the total number of actual purchases. Moreover, certain individuals post multiple reviews with similar content or exert significant influence over other reviewers. Deduplicating online reviews based on reviewer identification and incorporating detailed reviewer information, such as age, gender, and geographic location, are promising directions for future research to mitigate this issue. In particular, demographic data can help ensure that the insights derived from the RLA-IOA are not biased toward specific user groups that are more active in online environments.

Furthermore, determining the scope of analysis and the timespan of each review group remains a critical issue for future research. It is essential to divide review groups by comprehensively considering multiple factors, including product-release cycles and average lifespans. In this study, the scope of analysis and timespan for the review groups were set to three years and one year, respectively, based on the average smartphone lifecycle and typical remanufactured-product release timing. However, these parameters can be adjusted according to the market conditions and circumstances of the decision-makers. Future research should develop methodologies to determine the optimal scope and timespan for analysis.

Although the RLA-IOA framework supports designers in prioritizing and establishing design-for-circularity strategies, it does not address the underlying causes of obsolescence, such as technological, physical, or functional issues. Integrating cause analysis into the RLA-IOA could transform the framework into a more prescriptive review analytics tool, enabling companies to identify the root causes of obsolescence and adopt increasingly targeted and effective design-for-circularity strategies. Additionally, incorporating design optimization into the RLA-IOA can further advance the framework, allowing it to directly suggest optimal design solutions for circularity.

CRediT authorship contribution statement

Jisoo Won: Writing – original draft, Visualization, Validation, Methodology, Data curation. **Jiyeong Son:** Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Minjung Kwak:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT in order to improve readability and language of the work. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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